DIVISION OF NAVOTAS CITY

Senior High School

Statistics and Probability

(Quarter 1 and 2)

1st Semester

ADM
ALTERNATIVE DELIVERY MODE

S.Y. 2021-2022

NAVOTAS CITY PHILIPPINES

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Statistics and Probability for Senior High School
Alternative Delivery Mode
1st Semester (Quarter 1 and 2)
Second Edition, 2021

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Table of Contents

What I Know 1

Quarter I

Module 1.....4

Module 2.....11

Module 3.....19

Module 4.....25

Module 5.....32

Module 6.....39

Module 7.....43

Module 8.....48

Module 9.....51

Module 10.....57

Quarter II

Module 1.....63

Module 2.....72

Module 3.....77

Module 4.....80

Module 5.....86

Module 6.....89

Assessment.....93

Answer Key 94

References 96

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What I Know

Directions: Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

Quarter I

1. Is a numerical quantity that is assigned to the outcome of an experiment.
   A. Random variable
   B. Sample point
   C. Sample space
   D. Variable
2. In how many ways can two coins fall?
   A. 2
   B. 4
   C. 6
   D. 8
3. It tells the distance of score from the mean measured in standard deviation units.
   A. normal curve
   B. sample mean
   C. z-score
   D. area
4. Which of the following shows the probability that the z-score lies above a z-score value?
   $A. P(a < z < b)$
   $B. P(z > a)$
   C. $P(z < a)$
   D. $P(a = z$
5. What is the proportion of the area to the right of z = -1?
   A. -0.3413
   B. -0.8413
   C. 0.3413
   D. 0.8413

6. **Statement 1**: The number of students who are present in Filemon T. Lizan SHS for the first day of class for the S.Y. 2020-2021
   Statement 2: The number of Mayors in NCR who are present during the meeting
   Which of the following is CORRECT?

A. both statements are Discrete Random Variables
B. both statements are Continuous Random Variables
C. Statement 1 is Discrete Random Variable while the Statement 2 is Continuous Random Variables
D. Statement 1 is a Continuous Random Variable while the Statement 2 is a Continuous Random Variable

7. Statement 1: the volume of soft drinks in a 12-ounce can
   Statement 2: the time required to perform a job.
   which of the following is CORRECT?

A. both statements are Discrete Random Variables
B. both statements are Continuous Random Variables
C. Statement 1 is Discrete Random Variable while the Statement 2 is Continuous Random Variables
D. Statement 1 is a Continuous Random Variable while the Statement 2 is a Continuous Random Variable

8. Let B number of boys in a family and G for the girls in a family of four children. Determine the values of the random variable B.

   A. 0, 1
   B. 0, 1, 2
   C. 0, 1, 2, 3
   D. 0, 1, 2, 3, 4

For numbers 9 – 10. Consider the probability distribution of the number of mangoes given below.

R	3	2	1	0
P(R)	1/8	3/8	3/8	1/8

9. Find P(R = 3)

A. 1/8
B. 5/8
C. 3/8
D. 1

10. Find P(R > 1)

A. 18
B. 3/8
C. 1/2
D. 1

Quarter II

11. The random sample size n = 3 are drawn from a finite population consisting of the numbers 14, 25, 36, 47, 58 and 69. How many possible samples are there?

A. 12
B. 16
C. 20
D. 24

12. The random samples of size 3 are taken from a population of the numbers 1, 2, 3, 4, 5, 6, and 7. How many samples are there?

A. 35
B. 120
C. 210
D. 350

13. The random samples of size 4 are taken from a population of the numbers 1, 2, 3, 4, 5, 6, 7, and 8. How many samples are there?

A. 70
B. 840
C. 1 680
D. 3 024

14. The random sample size n = 5 are drawn from a finite population consisting of the numbers 15, 16, 17, 18, and 19. How many possible samples are there?

A. 1
B. 2
C. 4
D. 8

15. The following are the weights of five students in kg. suppose samples of size 2 are taken from this population of five students.

Student	Weight (in kg.)
Rusty	55
Buchoy	38
Boyong	60
Jenny	45
Kathrina	75

How many samples are possible?

A. 8
B. 10
C. 12
D. 14

16. A random sample of size 4 is taken with replacement from a population with μ = 12 and σ^2 = 8. Find the variance ($\sigma^2\bar{x}$).

A. 2.5
B. 2
C. 1
D. 1.5

17. A random of size 25 is taken with replacement from a population with μ = 121.4 and σ^2 = 50.5. Find the mean $\mu_{\bar{x}}$.

A. 121.4
B. 121.5
C. 122.5
D. 122

18. Which of the following is stated in the Central Limit Theorem?

A. The sampling distribution of the sample means approaches a normal distribution as the sample size decreases, no matter what the shape of the distribution is.
B. The sampling distribution of the sample means approaches a normal distribution as the sample size increases, no matter what the shape of the distribution is.
C. The sampling distribution of the sample means deviates from a normal distribution as the sample size decreases, no matter what the shape of the distribution is.
D. The sampling distribution of the sample means deviates from a normal distribution as the sample size increases, no matter what the shape of the distribution is.

19. Why is it important to sample not more than 10% of the population when the sample is drawn without replacement?

A. To reduce the effort when gathering data from the sample
B. To lessen the expenses that may occur during the conduct of research
C. To minimize the chance of creating a significant change when removing each item in the observation
D. To clearly see the behavior of the sample and then create a significant change from the population

20. The independence condition for the Central Limit Theorem is assumed to be met when _____.

A. the sample is biased
B. the sample is randomly selected
C. the sample is drawn with replacement
D. the sample is drawn without replacement

Quarter I

MODULE 1

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Statistics and Probability. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module consists of the lessons, namely:

– Understanding Random Variables
– Discrete Probability Distributions

After going through this module, you are expected to:

1. illustrates a random variable (discrete and continuous).
2. distinguishes between a discrete and a continuous random variable.
3. finds the possible values of a random variable.
4. illustrates a probability distribution for a discrete random variable and its properties.

Lesson 1	Random Variables and Probability Distributions

The concept of probability distribution is very important in statistical analysis of data. This is especially true when we try to estimate the true value of a variable, using sample data. Moreover, we also use probability distribution in testing hypothesis. We have studied probability in the previous grade levels. We have also studied frequency distributions in statistics. In this lesson you will learn to illustrate a random variable (discrete and continuous), distinguish between a discrete and a continuous random variable, find the possible values of a random variable, and illustrate a probability distribution for a discrete random variable and its properties.

What is It

RANDOM VARIABLE
- is a variable that assumes numerical values associated with the outcome of a random process or experiment.

OTHER DEFINITION OF TERMS
Experiment- any activity which can be done repeatedly under similar conditions.
Sample Space - the set of all possible outcomes in an experiment.
Event - a subset of a sample space.
Sample Point - the elements in a sample space**.**
Probability - the ratio of the number of favorable outcomes to the total number of possible outcomes.

A random variable may be classified as ***discrete*** or ***continuous.***

Discrete Random Variable - is one that can assume only a countable number of values.
Continuous Random Variable – can assume infinite number of values in one or more intervals.

Examples:
A. Classify the ff. if it is Discrete Random Variable or Continuous Random Variable.

1. number of pencils in a box	***Discrete Random Variable***
2. number of defective flashlights	***Discrete Random Variable***
3. voltage of radio batteries	***Continuous Random Variable***
4. amount of antibiotic in a vial	***Continuous Random Variable***
5. number of soldiers in a troop	***Discrete Random Variable***
6. length of wire ropes	***Continuous Random Variable***

Let's apply in a Problem Solving

Example 1

Supposed two coins are tossed and we are interested to determine the number of heads that will come out. Let us use H to represent the number of heads that will come out. Determine the values of the random variable H.

Let's follow the steps in solving this problem.

Step 1. List the sample space of the experiment.

$$S = \{ HH, HT, TH, TT \}$$

Step 2. Count the number of heads in each outcome and assign this number to this outcome.

Outcomes	Number of Heads (Value of H)
HH	2
HT	1
TH	1
TT	0

The values of the **random variable H** (no. of heads) in this experiment are 0, 1, and 2.

Example 2

A basket contains 10 ripe and 4 unripe mangoes. If three mangoes are taken from the basket one after the other, determine the possible values of the random variable R representing the number of ripe mangoes.

Solution: let **R** represents ripe mangoes and **U** represents unripe mangoes

Step 1. List the sample space of the experiment.

S = {RRR, RRU, RUR, URR, UUR, URU, RUU, UUU}

Step 2. Count the number of ripe mangoes (R) in each outcome and assign this number to this outcome.

Outcome	No. of ripe mangoes (Value of R)
RRR	3
RRU	2
RUR	2
URR	2
UUR	1
URU	1
RUU	1
UUU	0

The values of the random variable r (number of ripe mangoes) in this experiment are 0, 1, 2, and 3.

Different Presentations of a Discrete Probability Distribution

Probability Distribution of a discrete random variable – is a correspondence that assigns probabilities to the values of a random variable. The probability distribution of a discrete random variable is also called the **probability mass function.**

The probability distribution of a discrete random variable can be shown graphically by constructing a histogram. The graph is called a **probability histogram.**

Probability Distribution of the Number of Ripe Mangoes

R	**3**	**2**	**1**	**0**
P(R)	**1/8**	**3/8**	**3/8**	**1/8**

For any discrete random variable X, the following are true.

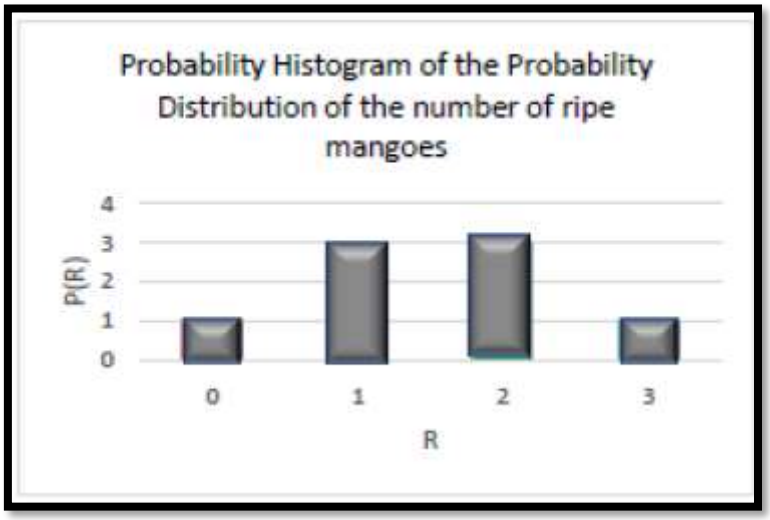
- **0 ≤ P(X) ≤ 1**, for each value of X
- **ΣP(X) = 1**

So, if we add the P(R) values that is equal to 1.

P(R) = $\frac{1}{8}+\frac{3}{8}+\frac{3}{8}+\frac{1}{8}=\frac{8}{8}$

= $\frac{1}{8}$ *or* 1

Therefore, it is a probability distribution of a discrete random variable and sometimes it is called probability mass function.



The bar graph shows the relationships of R which is value of the random variables and the P(R) which is the probability of the number of ripe mangoes.

If we continue the process…

Step 3. Construct the frequency distribution of the values of the random variable R.

Number of Ripe Mangoes (Values of R)	**Number of Occurrence (frequency)**
3	1
2	3
1	3
0	1
Total	8

Step 4. Construct the probability distribution of the random variable R by getting the probability of occurrence of each value of the random variable.

Number of Ripe Mangoes (Values of R)	**Number of Occurrence (frequency)**	**Probability P(R)**
3	1	1/8
2	3	3/8
1	3	3/8
0	1	1/8
Total	**8**	**1**

The probability distribution of the random variable R can be written as follows:

R	**3**	**2**	**1**	**0**
P(R)	**1/8**	**3/8**	**3/8**	**1/8**

Properties of a Discrete Probability Distribution

Examine the probability distribution that we have learned in the given example. What have you notice about the probability values of the random variable in each probability distribution?

What is the sum of the probabilities of a random variable?
Consider the probability of the number of bananas given below.

R	**3**	**2**	**1**	**0**
P(R)	**1/8**	**3/8**	**3/8**	**1/8**

Solve the following:

1. P (R = 2)

In words, the probability of R which is exactly 2.

Solution: Since the given is exactly 2

Therefore, the answer is **3/8**.

2. P (R = 3)

In words, the probability of R which is exactly 3.

Solution: Since the given is exactly 3

Therefore, the answer is **1/8**.

3. P (R > 1)

In words, the probability of R which is greater than 1.

There are two possible values of R. These are 2 and 3.

P (R > 1) = P(2) + P(3)

= 3/8 + 1/8

= 4/8 or ½

Note: Simplify the answer if possible

4. P (R < 3)

In words, the probability of R which is less than 3.

There are three possible values of R. These are 2, 1 and 0.

P (R < 3) = P (2) + P(1) + P(0)

= 3/8 + 3/8 + 1/8

= 7/8

5. ΣP(R)

To find ΣP(R) we need to find the sum of all the probability values.

ΣP(R) = P (3) + P (2) + P (1) + P(0)

= 1/8 + 3/8 + 3/8 +1/8

= 8/8 or 1

What's More

Classify Me Please!

Directions: Classify the following as *discrete random variable* (DRV) or *continuous random variable* (CRV). Write your answers on a separate sheet of paper.

Statement
1. the number of senators present in the meeting
2. the weight of newborn babies for the month of June
3. the number of ballpens in the box
4. the capacity of electrical resistors
5. the amount of salt needed to bake a loaf of bread

6. the capacity of an auditorium
7. the number of households with television
8. the height of mango tree in a farm
9. the area of lots in a subdivision
10. the number of students who joined the fieldtrip
11. the number of children in a family
12. the number of tails flipped in 4 trials
13. the time required to perform a job
14. the amount of sugar in a pineapple juice
15. the volume of mango juice in a 12 – ounce can
16. the Saturday night attendance at the prayer meeting
17. the number of patients of Dr. Naval in his clinic for three weeks
18. the time taken to complete an examination in Statistics and Probability
19. the interest rate given by the BDO bank
20. the weight of a fish

What I Can Do

Analyze Me Please!

Directions: Determine the values of the random variable in each of the following situations. Write your answers on a separate sheet of paper.

1. A coin is flipped four times. Let T be the number of tails that come out. Determine the values of the random variable T.

a. List the sample space of the experiment.

S = { _____}

b. Count the number of tails (T) in each outcome and assign this number to this outcome.

Outcome	Number of Tails (Value of T)

c. The values of the random variables are_____

2. A box contains 4 green and 2 blue dice. Three dice are chosen one after the other. Determine the values of the random variable G representing the number of green dice.

a. List the sample space of the experiment.

S = { _____}

b. Count the number of green dice (G) in each outcome and assign this number to this outcome.

Outcome	Number of Tails (Value of T)

c. The values of the random variables are_____

3. A meeting of consuls was attended by 4 Americans and 2 Germans. If three consuls were selected at random, construct the probability distribution for the random variable G representing the number of Germans.

a. List the sample space of the experiment.

S = { _____}

b. Count the number of Germans (G) in each outcome and assign this number to this outcome.

Outcome	Number of Germans (Value of G)

c. The values of the random variables are_____

MODULE 2

This module was designed and written with you in mind. It is here to help you master the random variable and probability distributions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into two lessons, namely:

- Lesson 1 – Constructing Probability Distribution
- Lesson 2 – Mean, Variance, and Standard Deviation of a Discrete Random Variable

After going through this module, you are expected to:

1. computes probabilities corresponding to a given random variable
2. illustrates the mean and variance of a discrete random variable.
3. calculates the mean and the variance of a discrete random variable

Lesson 2.1 Constructing Probability Distributions

In your previous study of mathematics, you have learned how to find the probability of an event. In this lesson, you will learn how to construct a probability distribution of a discrete random variable. Your knowledge of getting the probability of an event is very important in understanding the present lesson. To find out if you are ready to learn this new lesson, do the following activities.

What is It

Discrete probability distribution Defined

A discrete probability distribution is a table showing all the possible values of a discrete random variable together with their corresponding probabilities.

The mean of a discrete random variable x is also called the **expected value of x**. It is the weighted average of all the values that the random variable x would assume in the long run. The discrete random variable x assumes values or outcomes in every trial of an experiment with their corresponding probabilities. The expected value of x is the average of the outcomes that is likely to be obtained if the trials are repeated. The expected value of x is denoted by E(x).

The mean expected value of a discrete random variable X is computed using the following formula: $E(X)=\Sigma[xP(x)]$

Where X = discrete random variable

X = outcome or value of the random variable

P(x) = probability of the outcome x

A probability distribution describes the probability of each specific value in a random variable. The probability distribution of a discrete random variable is probability mass function (pmf). The pmf of x is denoted by $f(x)$ and satisfies the following two basic properties.

a. $f(x) = P(X = x) \geq 0 \; if \; x \in the\ Support\ S$ or simply, sample space

b. $\sum_{x \in S} f(x) = \sum_{x \in S} P(X = x) = 1$

According to the first property, for every element x in the Support S, in another words, sample space, all the probabilities must be positive and according to the second property, the sum of all the probabilities for all possible x values in the Support S must be equal to 1. The values of the discrete random variable X where $\boldsymbol{f(x)>0}$ are called its mass points.

> The Support S or the Sample Space of a random variable is the set of values that the random variable can take. It contains countably infinite number of possible values. This means that the elements of S can be put into one- to -one correspondence with the set of natural numbers.

Example No.1 : NUMBER OF TAILS

Suppose three coins are tossed. Let **Y** be the **random variable** representing the **number of tails** that occur. Find the probability of each of the values of the random variable Y.

Solution:

<table>
<tr><th>Steps</th><th>Solution</th></tr>
<tr><td>1. Determine the sample space. Let H represent head and T represent Tail</td><td>The sample space for this experiment is : S={$TTT, TTH, THT, HTT, HHT, HTH, THH, HHH$}</td></tr>
<tr><td>2. Count the number of tails in each outcome in the sample space and assign this number to this outcome</td><td><table><tr><th>Possible Outcomes</th><th>Value of the Random Variable Y(No. of Tails)</th></tr><tr><td>TTT</td><td>3</td></tr><tr><td>TTH</td><td>2</td></tr><tr><td>THT</td><td>2</td></tr><tr><td>HTT</td><td>2</td></tr><tr><td>HHT</td><td>1</td></tr><tr><td>HTH</td><td>1</td></tr><tr><td>THH</td><td>1</td></tr><tr><td>HHH</td><td>0</td></tr></table></td></tr>
<tr><td>3. There are four possible values of the random variable Y, representing the number of tails. These are 0,1,2 and 3. Assign probability values P (Y) to each value of the random variable
-There are 8 possible outcomes, and no tail occurs once, so the probability that we shall assign to the random 0 is $\frac{1}{8}$
-There are 8 possible outcomes and 1 tail occurs three times, so the probability that we shall assign to the random variable 1 is $\frac{3}{8}$
-There are 8 possible outcomes and 2 tail occurs three times, so the probability that we shall assign to the random variable 2 is $\frac{3}{8}$
-There are 8 possible outcomes and 3 tail occurs once
-e, so the probability that we shall assign to the random variable 2 is $\frac{1}{8}$</td><td><table><tr><th>Number of Tails (Y)</th><th>Possibility P(Y)</th></tr><tr><td>0</td><td>$\frac{1}{8}$</td></tr><tr><td>1</td><td>$\frac{3}{8}$</td></tr><tr><td>2</td><td>$\frac{3}{8}$</td></tr><tr><td>3</td><td>$\frac{1}{8}$</td></tr></table></td></tr>
</table>

Table 1.1 . The Probability Distribution or the Probability Mass Function of Discrete Random Variable Y.

No. of Tails (Y)	0	1	2	3
Probability P (Y)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

The sum of probabilities is $\sum \boldsymbol{P(Y)} = \mathbf{1}$. **This is discrete random variable**

Example 2. NUMBER OF BLUE BALLS

Two balls are drawn in succession without replacement from an URN containing 5 red balls and 6 blue balls. Let **Z** be the **random variable** representing the **number of blue balls**. Construct the probability distribution of the random variable Z.

Solution:

<table>
<tr><th>Steps</th><th>Solution</th></tr>
<tr><td>1. Determine the sample space. Let B represent the blue ball and R represent the red ball.</td><td>The sample space for this experiment is :
S={RR, RB, BR, BB}</td></tr>
<tr><td>2. Count the number of blue balls in each outcome in the sample space and assign this number to this outcome</td><td><table><tr><th>Possible Outcomes</th><th>Value of the Random Variable Z(No. of blue balls)</th></tr><tr><td>BB</td><td>2</td></tr><tr><td>BR</td><td>1</td></tr><tr><td>RB</td><td>1</td></tr><tr><td>RR</td><td>0</td></tr></table></td></tr>
<tr><td>3. There are three possible values of the random variable Z, representing the number of blue balls. These are 0,1, and 2. Assign probability values P (Z) to each value of the random variable

-There are 4 possible outcomes, and no blue balls occurs once, so the probability that we shall assign to the random 0 is $\frac{1}{4}$

-There are 4 possible outcomes, and 1 blue ball occurs two times, so the probability that we shall assign to the random variable 1 is $\frac{2}{4}$ or $\frac{1}{2}$

-There are 4 possible outcomes, and 2 blue balls occurs once, so the probability that we shall assign to the random variable 2 is $\frac{1}{4}$</td><td><table><tr><th>Number of Tails (Y)</th><th>Possibility P(Y)</th></tr><tr><td>0</td><td>$\frac{1}{4}$</td></tr><tr><td>1</td><td>$\frac{1}{2}$</td></tr><tr><td>2</td><td>$\frac{1}{4}$</td></tr></table></td></tr>
</table>

Table 1.2 . The Probability Distribution or the Probability Mass Function of Discrete Random Variable Z.

No. of blue balls (Z)	0	1	2
Probability P (Z)	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

The sum of probabilities is $\sum \boldsymbol{P(z)} = \mathbf{1}$. **Discrete the random variable because the sum of probabilities is equal to 1.**

Example 3. Determine whether the given values can serve as the values of a probability distribution of the random variable x that can take on only the values 1,2,3,4. P(1) $=\frac{1}{19}$, P(2)= $\frac{10}{19}$, P (3)= $\frac{5}{19}$, P(4) =)= $\frac{5}{19}$.

P	1	2	3	4
P(x)	$\frac{1}{19}$	$\frac{10}{19}$	$\frac{5}{19}$	$\frac{5}{19}$

The sum of probabilities is $\sum \boldsymbol{P(z)} = \mathbf{1.163}$ **, this is not a discrete random variable because the sum of probabilities is not equal to 1.**

Example 4. Determine whether the given values can serve as the values of a probability distribution of the random variable X. P(x) = $\frac{1}{8}$ for x= 1,2,3…8.

Solution :

x	1	2	3	4	5	6	7	8
P(x)	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

The sum of probabilities is $\sum \boldsymbol{P(x)} = \mathbf{1}$**, this is a discrete random variable because the sum of probabilities is equal to 1.**

Lesson 2.2 Mean, Variance and Standard Deviation of a Discrete Random Variable

The **mean or expected value** of a discrete random variable x is computed using the following formula:

E(x)=$\Sigma[xP(x)]$, where x = discrete random variable

x = outcome or value of the random variable

P (x) = probability of the outcome x

The variance of a random variable X is denoted by $\sigma 2$. It can likewise be written as Var (X). The variance of a random variable is the expected value of the square of the difference between the assumed value of random variable and the mean. The variance of X is:

$$Var(X) = \Sigma[(x-\mu)2P(x)] \text{ or } \sigma 2 = \Sigma[(x-\mu)2P(x)]$$

Where:

x = outcome, μ= $population\ mean$, P (x) = probability of the outcome

The larger the value of the variance, the farther are the values of X from the mean. The variance is tricky to interpret since it uses the square of the unit of measure of x. so, it is easier to interpret the value of the standard deviation because it uses the same unit of measure of x.

The standard deviation of a discrete random variable x is written as σ. It is the square roots of the variance. The standard deviation is computed as:

$$\sigma = \sqrt{\Sigma[(x-\mu)2)P(x)]}$$

Variable x represents the number of college graduates in the households. The probability distribution of x is shown below:

x	**0**	**1**	**2**
P(x)	0.25	0.50	0.25

Find the mean or expected value of x.

x	**P(x)**	**xP(x)**
0	**0.25**	0
1	**0.50**	0.50
2	**0.25**	0.50
		$\Sigma[xP(x)]$= 1

E(x)= $\Sigma[xP(x)]$ =1.00

The expected value is 1. So, the average number of college graduates in the household of the small town is one.

Example 2. A security guard recorded the number of people entering the bank every hour during one working day. The random variable x represents the number of people who entered the bank. The probability distribution of x is shown below.

x	**0**	**1**	**2**	**3**	**4**	**5**
P(x)	0	0.1	0.2	0.4	0.2	0.1

What is the expected number of people who enters the bank every hour?
Solution:

x	P(x)	xP(x)
0	**0**	0
1	**0.1**	0.1
2	**0.2**	0.4
3	**0.4**	1.2
4	**0.2**	0.8
5	**0.1**	0.5
	$\Sigma P(x)$= 1	**$\Sigma[xP(x)]$= 3**

So, E(x) = 3.0

Therefore, the average number of people entering the bank every hour during that working day is three.

Example 3. Determine the variance and the standard deviation of the following probability mass function.

x	**1**	**2**	**3**	**4**	**5**	**6**
P(x)	0.15	0.25	0.30	0.15	0.10	0.05

Solution:

Steps

1. Find the expected value.
2. Subtract the expected value from each outcome. Square each difference
3. Multiply each difference by the corresponding probability
4. Sum up all the figures obtained in step 3.

x	P(x)	xP(x)	$x-\mu$	$(x-\mu)^2$	$(x-\mu)^2P(x)$
1	0.15	0.15	1 – 2.95=-1.95	3.8025	0.570375
2	0.25	0.50	2-2.95=-0.95	0.9025	0.225625
3	0.30	0.90	3-2.95=0.05	0.0025	0.000750
4	0.15	0.60	4-2.95=1.05	1.1025	0.165375
5	0.10	0.50	5-2.95=2.05	4.2025	0.420250
6	0.05	0.30	6-2.95=3.05	9.3025	0.465125
		$\Sigma[xP(x)]$=2.95	$\Sigma[(x-\mu)^2P\ (2)]$ =1.8475		

E(x) = $\Sigma[xP(x)]$ =2.95

$\sigma 2$ = $\Sigma[(x-\mu)^2P\ (2)]$ =1.8475

$\sigma=\sqrt{\Sigma[(x-\mu)^2)\ P(x)]}$

$=\sqrt{1.8475}$ = 1.359227 or 1.36

What's More

A. Construct a probability distribution for the data. (2 points)

1. The probabilities that a surgeon operates on 3,4,5,6, or 7 patients in any one day are 0.15,0.20 ,0.25,0.20, and 0.20 respectively.
2. The probabilities that a customer buy 2,3,4,5, or 6 items in a convenience store are 0.32,0.12,0.23,0.18, and 0.15 respectively.
3. The probabilities that a student will borrow 1,2,3, or 4 books are 0.45,0.30,0.15, and 0.10, respectively.
4. The probabilities that a bias die will fall as 1,2,3,4,5 or 6 are $\frac{1}{2}, \frac{1}{6}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}$ and $\frac{1}{12}$,, respectively.
5. The probabilities that a dispositor will invest Php100,000, Php250,000., or Php180,000 are $\frac{1}{4}, \frac{1}{4}$,and$\frac{1}{4}$, respectively.

What I Can Do

A. Find the expected value of each probability mass function below. (2 points)

1.

x	0	1	2	3
P(x)	0.15	0.32	0.37	0.16

2.

x	0	1	2	3
P(x)	0.17	0.33	0.36	0.14

3.

x	0	1	2	3
P(x)	0.20	0.30	0.32	0.18

B. Find the variance and standard deviation of each of the following probability distribution. (3 points)

1.

X	0	1	2	3
P(x)	0.10	0.45	0.25	0.20

2.

X	0	1	2	3
P(x)	0.15	0.38	0.33	0.14

MODULE 3

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the random variable and probability distributions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module deals with an understanding of:

- The Mean of a Discrete Probability Distribution and the Variance of a Discrete Probability Distribution
- Applied Problems involving the Mean and the Variance of a Discrete Probability Distribution

After going through this module, you are expected to:

1. Interpret the mean and the variance of a discrete random variable
2. Solves problems involving mean and the variance of a probability distribution.

Lesson 3 Finding the Mean and the Variance of a Discrete Probability Distribution

In Lessons 1 and 2, you have learned and understood that the probability of distribution of a discrete random variable is also called the ***probability mass function,*** and a listing of the possible values of a discrete random variable is a ***probability distribution***.

In this lesson, you shall have to learn how to find the mean and the variance and also the standard deviation of a probability distribution.

What is It

The table below shows the result of 4 tiles picked and returned in the jar 15 times. If x represents each tile, and *f* represents the number of times picked, your task is to evaluate what is being asked you to do.

Tiles x	Number of times picked f	fx
1	2	2
2	4	8
3	8	24
4	1	4
	Total =15	

Let us analyze and explore.

Just like frequency distribution, the probability distribution can be described by computing its mean and variance. This time you will be exploring how to compute for the mean and the variance for the discrete probability distribution.

THE MEAN OF A DISCRETE PROBABILITY DISTRIBUTION

To find the mean (μ) or the expected value E(x) of a discrete probability distribution, we use the following formula:

$$\mu = \mathbf{E}(x) = \sum [x\, P(x)]$$

where: μ = mean

x = value of the random variable

P(x)= is the probability value of the random variable

From the experiment we discussed on tile and jar, we can use x to represent the *tiled number* and the *number of times picked* to P(x) and dividing each of its value by 15, thus table becomes:

x	1	2	3	4
P(x)	2	4	8	1

<u>Let us figure it out:</u>

(1). Find the mean of the discrete random variable χ using the table above.

Step 1: Multiply the value of x by its corresponding probability value P(x).

x	P(x)	xP(x)
1	$\frac{2}{15}$	$\frac{2}{15}$
2	$\frac{4}{15}$	$\frac{8}{15}$
3	$\frac{8}{15}$	$\frac{24}{45}$
4	$\frac{1}{15}$	$\frac{4}{15}$
		$\sum[x\ \mathrm{P}(x)] = \frac{38}{15}$

Step 2: Find the mean or the expected value of the probability distribution by getting the sum of the values under the column x P(x).

$$\mu = \mathrm{E}\ (\mathrm{x}) = \sum[xP(x)]$$

$$= \frac{38}{15}$$

$$= \mathbf{2.53}$$

therefore, the mean or the expected value of the probability distribution is **2.53.**

THE VARIANCE AND STANDARD DEVIATION OF THE DISCRETE PROBABILITY DISTRIBUTION

To find the variance and the standard deviation of a discrete probability distribution using these formulas:

$$\boldsymbol{\sigma^2 = \sum[(x^2\ P(x))] - \mu^2} \qquad \boldsymbol{\sigma = \sqrt{\sum[(x^2\ P(x))] - \mu^2}}$$

where: *μ = mean*

X = value of the random variable

P(X)= the probability value of the random variables

σ^2 = variance

σ = standard deviation or SD

Now, let us try to find the variance and the standard deviation of the discrete random variable x using the same example we use.

x	P(x)
1	$\frac{2}{15}$
2	$\frac{4}{15}$
3	$\frac{8}{15}$
4	$\frac{1}{15}$

Step 1: Find the mean of the probability distribution. Prepare a table as shown below.

x	P(x)	xP(x)
1	$\frac{2}{15}$	$\frac{2}{15}$
2	$\frac{4}{15}$	$\frac{8}{15}$
3	$\frac{8}{15}$	$\frac{24}{45}$
4	$\frac{1}{15}$	$\frac{4}{15}$
		$\sum[x\,\mathrm{P}(x)] = \frac{38}{15}$

Using the formula for the mean of the probability distribution:

$$\mu = \mathbf{E}\,(x) = \sum[(xP(x))]$$

$$= \frac{38}{15} = 2.53$$

Step 2: Square each value of the random variable and multiply by the corresponding probability value (x^2 P(x)). The new table below will give you an idea on how to do the squaring values of the random variable

x	x^2	P(x)	x^2 P(x)	xP(x)
1	1	$\frac{2}{15}$	$\frac{2}{15}$	$\frac{2}{15}$
2	4	$\frac{4}{15}$	$4\left(\frac{4}{15}\right)$	$\frac{8}{15}$
3	9	$\frac{8}{15}$	$9\left(\frac{2}{15}\right)$	$\frac{24}{45}$
4	16	$\frac{1}{15}$	$16\left(\frac{1}{15}\right)$	$\frac{4}{15}$
			$\sum[(x^2\,P(x))] = \frac{106}{15} \approx 7.06$	$\sum[x\,\mathrm{P}(x)] = \frac{38}{15} \approx 2.53\overline{3}$

Step 3: Find the variance and the standard deviation by applying the formulas

$$\sigma^2 = \sum[(x^2\,P(x))] - \mu^2$$

= 7.06 – 2.53

= 4.53

$$\sigma = \sqrt{\sum[(x^2\,P(x))] - \mu^2}$$

= 2.13

or $\sqrt{\sigma^2}$ = **2.13** thus, the variance is 4.53 and the standard deviation is 2.13

this only shows how close the variance and standard deviation from the mean. Since the experiment, we have discussed is a tile picked and returned in the jar 15 times. The mean computed tells us that for every tile picked from the jar, the number in the tile is in average.

What's More

Thinking-Out-Loud.

To give you a better understanding of the Mean, Variance, and Standard Deviation of a Discrete Probability Distribution. Try to answer the following by completing the table.

1.What is the ***mean*** outcome if a fair die is rolled?

Step 1: Since the die is fair then:

x	$P(x)$	$xP(x)$
1	$\frac{1}{6}$	
2	—	
3	—	
4	$\frac{1}{6}$	$4\left(\frac{1}{6}\right) = \frac{4}{6}$
5	—	
6	—	

(2) The random variable χ, representing the number of nuts in a chocolate bar has the following probability distribution. Compute the mean.

x	0	1	2	3	4
$P(x)$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{1}{10}$

(3). The probability distributions below show the number of typing errors (x) and the probability $P(x)$ of committing the errors whenever clerks' type-in a document. Complete the table.

x	0	1	2	3	4	5
$P(x)$	0.02	0.22	0.42	0.31	0.10	0.04
$x\,P(x)$						
x^2						
$x^2P(x)$						

Refer to the table in no. 3 to answer numbers 4 – 5.

(4) Compute the variance.

(5) Compute for the standard deviation.

What I Can Do

Let me check what you gain from the lesson.

1. Complete the table below and find the Mean or expected value of the following probability distribution.

x	0	1	2	3	4
P(x)	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
xP(x)					

2.

H	**P(H)**	**HP(H)**
0	0.06	
1	0.70	
2	0.20	
3	0.03	
4	0.01	

3-4. Determine the variance and the standard deviation of the random variable.

x	1	2	3	4	5
P(x)	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
xP(x)					

MODULE 4

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Statistics and Probability. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

After going through this module, you are expected to:

1. illustrate a normal random variable and its characteristics.
2. identify regions under the normal curve corresponding to different standard normal values.
3. convert a normal random variable to a standard normal variable and vice-versa.
4. compute probabilities and percentiles using the standard normal table; and
5. apply the normal curve concepts in solving problems.

Lesson 4	Exploring the Normal Curve Distribution

In the previous module, you have learned to solve problems involving the mean and variance of a probability distribution. However, the data you used previously are samples of discrete data. What if the data belong to a continuous type? This module will help you deal with problems involving distribution of this type. Traditionally, we call this a normal probability distribution or simply the normal curve

What is It

A. PROPERTIES OF THE NORMAL CURVE

Normal distribution or normal curve represents a group of data where a very large number of cases exists and the mean, the median and the mode are all equal. When you sketch the graph of a normal curve, you will find the following properties:

1. The distribution curve is unimodal and bell-shaped. Unimodal means that there is only one peak point.
2. The mean, the median, and the mode coincide at the center.

3. The curve is symmetrical about its center. Meaning, when you draw a vertical line at the center of the curve, the resulting half part looks an image of the other half part.
4. The width of the curve is based on the standard deviation of the distribution.
5. The tails of the curve approach the base line, but it will never intersect the line. These tails just go nearer and nearer to the base line, but never meet the line.
6. The area of the curve is 1. Thus, normal curve is also a probability distribution.

B. EXPLORING THE STANDARD NORMAL CURVE

The normal curve is a standard normal curve when the mean $\mu = 0$ and the standard deviation $\sigma = 1$. This is mostly used to represent inferential statistics. You will find its area by substituting the mean $\mu = 0$ and the standard deviation $\sigma = 1$ in the formula that describes a normal curve. But don't worry! Mathematicians have already computed these for everyone's use.

Look at the image below. This is a graphical representation of a normal cuve. distribution.

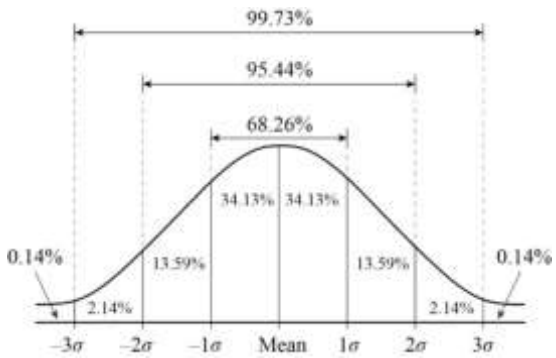


Fig 4.1. Areas under the Normal Curve

Source: Chegg Study. https://che.gg/2YDK2zM

You might be wondering why the area is considered as equal to 1 when the standard practice is to show 99.73% of the area. Take note that .9973 is just the area between -3 and +3. In this case, remember that the total area is not shown because the tails are asymptotic to the horizontal line. Meaning, it just continues to approach the line but will never intersect the line. Therefore, there is a little portion of the area at the tails of the distribution. So, when asked about the area under a normal curve, you say 1.

Areas under the normal curve is found at the z-Table. This time, you will learn how to use z-table in finding the areas under the normal curve.

Steps in Finding the Areas under the Normal Curve Given a Z-Value

1. Express the given z-value into a three-digit form.
2. Locate the first two digits on the left column of the z-Table.
3. Match the third digit with the appropriate column on the right just like in what you are doing in multiplication table.
4. The intersection of the row and the column is the required area or probability.

Illustrative Example 1: Find the area that corresponds to z = 0.72.

***Note**: The area that corresponds to z = 0.72 can also be understood as "the corresponding area between z = 0 and z = 0.72."*

Steps:

1. Express the z-value into a three-digit form.
   - z = 0.72 is already in three digits.
2. Locate the first two digits on the left column of the z-Table.
   - The first two digits are 0.7. Find it in the left column.
3. In the z-table, match the third digit with the appropriate column on the right just like in what you are doing in multiplication table.
   - The last digit is 2. Find the column with the heading .02.
4. The intersection of the row and the column is the required area or probability.
   - The area is 0.2642.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7			0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441

How will you show this in a graph?

- Sketch a normal curve.
- Draw a vertical line through the given z-values and shade the region.
- Note that the given z-value is positive, so the shaded region is on the right of the mean

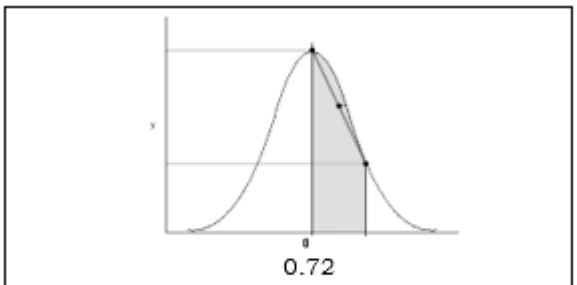
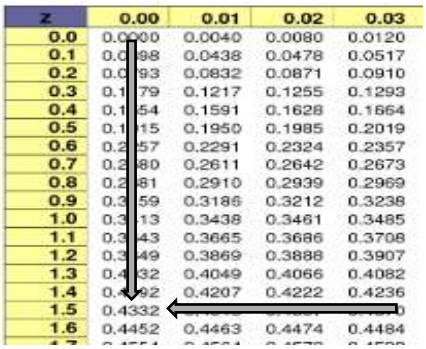


Figure 4.2. Region under the Normal Curve

Source: StatisticsHowTo.com

Illustrative Example 2: Find the area that corresponds to z = -1.5.

Notes: The area that corresponds to z = -1.5 can also be understood as "the corresponding area between z = 0 and z = -1.5." Moreover, note that "negative sign" in the z-value is just a signal that the region is on the left side of the mean. This means that the area corresponding to z = 1.5 is also the same with z = -1.5. The only difference is their location on the graph. If it is positive, then the region is on the right of the mean. If it is negative, then the region is on the left of the mean.

Steps	Solution
Express the z-value into a three-digit form.	z = -1.50
Locate the first two digits on the left column of the z-Table.	
In the z-Table, match the third digit with the appropriate column on the right	
Find the intersection of the row 1.5 and the column .00	z = .4332 Therefore, the required area is .4332

Now, sketch a normal curve and identify the region. Remember that the z-value is negative, so the region is on the left of the mean.

Before proceeding to the next topic, remember that the mean divides the area under the curve

C. UNDERSTANDING THE Z-SCORES

Recall that z-value or z-score tells you the distance from the mean measured in standard deviation units. It can be positive (above the mean), negative (below the mean), or zero (equal to the mean). However, in real life, these scores are not usually given. Thus, it is important that you know how to transform a raw score to its corresponding z-score under the normal curve.

To get the z-value, use the formula:

$$z = \frac{X-\mu}{\sigma} \quad (z - score\ for\ population\ data)$$

OR

$$z = \frac{X-\bar{X}}{s} \quad (z - score\ for\ sample\ data)$$

where: $x = given\ measurement$

$\mu = population\ mean$

$\sigma = population\ standard\ deviation$

$\bar{x} = sample\ mean$

$s = sample\ standard\ deviation$

Example 1: Find the z-value that corresponds to a score X = 58, given the mean μ = 50 , and standard deviation σ = 4.

(*The symbols used are for population. Therefore, the z-score locates the raw score within a population.*) Solution:

$$z = \frac{X-\mu}{\sigma} = \frac{58-50}{4} = \frac{8}{4} = 2$$ (*The resulting z-value is* ***positive*** *2.*)

The corresponding z-score to the raw score 58 is 2.

Meaning, the score 58 is 2 units **above** the mean.

Example 2: Locate the corresponding z-value to a score of 20 given that $\bar{X}$ = 26 and s = 4. (*The given are sample data.*)

Solution:

$$z = \frac{X-\bar{X}}{s} = \frac{20-26}{4} = \frac{-6}{4} = -1.5$$ (*The resulting z-value is* ***negative*** *1.5)*

The corresponding z-score to the raw score 20 is -1.5.

Meaning, the score 20 is 1.5 units **below** the mean.

Example 3: During the summer break, 1500 students took a test to apply for scholarships in college. Their mean score is 80 with a standard deviation of 5. How many students got a score between 75 and 82?

Steps	Solution
1. Convert the raw scores 75 and 82 to z-scores.	For the raw score 75, $z = \frac{X-\mu}{\sigma} = \frac{75-80}{5} = \frac{-5}{5}$ =-1 For the raw score 82, $z = \frac{X-\mu}{\sigma} = \frac{82-80}{5} = \frac{2}{5} = 0.4$
2. Find the area that corresponds to each z-score.	z = -1 corresponds to the area .3413 z = .4 corresponds to the area .1554
3. Sketch the graph of a normal curve showing the z-scores	
4. In your sketch, draw a line through the z-scores and shade the region between them.	
5. Analyze the sketch and determine the operation to use to find the total area.	The graph suggests addition .3413 + .1554 =.4967 .4967 is 49.67% when expressed to percent.
6. Make a statement.	49.67% of the students got a score between 75 and 82.

D. LOCATING PERCENTILES UNDER THE NORMAL CURVE

The following phrases are expressions of order. Are you familiar with them?

"Top 10"

'a score of 75%'

Just like the z-scores, percentile tells the position of a value. It describes the relationship of a value to the rest of the data. It is a point in the distribution where a number of cases is **below** it. For example, if your score is at the 84th percentile, it means that 84% of the scores were lower than yours and that 16% of the scores were higher than yours.

Example: Locate the P_{93} in the normal curve.

Steps	Solutions
1. Sketch the appropriate normal curve.	
2. Write the given percentage as probability.	93% is the same as 0.9300 or .9300
3. Split .9300 into .5000 and .4300.	.9300 = .5000 + .4300
4. Shade .5000 of the sketch of the normal curve in Step 1.	
5. Look at the z-Table and find the area .4300 in the body of the table. Did you it?	This area is not found in the table. But you can see that it is between .4292 and .4306
6. Find the z-score that corresponds to the two nearest areas to .4300.	.4300 is between .4292 and .4306
7. Find the z-value that corresponds to .4292	z = 1.47
8. Find the z-value that corresponds to .4306	z = 1.48
9. Find the average of the two z-values.	$z = \frac{1.47 + 1.48}{2} = \frac{2.95}{2} = 1.475$
10. Go back to your sketch in step 1 and locate (*just estimate*) z = 1.475. Then, draw a vertical line through this point. Finally, shade the region **to the left** of z = 1.475.	The shaded region is 95% of the distribution.

What's More

The Neophyte Statistician

Hello, neophyte! Prove your learning by solving the following problems.

1,000 children joined the physical fitness program last month. Their average weight before the program was 35 kg with a standard deviation of 5 kg. How many of these children weighted between 32 kg and 45 kg?

Write your solutions in the figure.

What I Can Do

1. The scores in a test conducted to 500 students are normally distributed with a mean of 110 and a standard deviation of 6. How many students got a score below 104?

2. 150 male students were asked about their shoe size. The result is normally distributed with a mean 8 inches and a standard deviation 3 inches. Find the probability that a male student, picked at random, has a shoe size 8 in.

3. The heights of 800 children is normally distributed with an average 95 cm and standard deviation 5 cm. How many of the children has a height between 93 cm and 98 cm?

MODULE 5

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Statistics and Probability. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is concentrated on the lesson

- Lesson 5 – Understanding Sampling and Sampling Distribution

After going through this module, you are expected to:

1. illustrates random sampling.
2. distinguishes between parameter and statistic.
3. Identifies sampling distributions of statistics (sample mean).

Lesson 5 Understanding Sampling and Sampling Distribution

In the first lesson of this module, we have learned how to construct the probability distribution of a discrete random variable. We have also learned how to compute the mean and the standard deviation of a discrete random variable. We have also studied to identify regions under the normal curve, to convert a normal random variable to a standard normal variable and vice-versa, and to compute probabilities and percentiles using the standard normal table. In this lesson we will learn to illustrate random sampling, to distinguish between parameter and statistic and to identify sampling distributions of statistics (sample mean). Also, we will learn how to construct the sampling distribution of sample means and find out some characteristics of the sampling distribution of the sample means. This will eventually help us to understand the process of making statistical inference about the population, using a sample drawn from it.

What is It

Sample - a subset of the population from which the data is collected. It is a small part of the population from which the researchers gather data.

OTHER DEFINITION OF TERMS

Sampling method-It is concerned with selecting a subset of the population used to estimate the entire population's characteristics. **Random Sampling** is a method wherein each element of the population has an equal chance of being chosen to represent the population.

TYPES OF RANDOM SAMPLING

a. Simple random sampling - is the simplest form of random sampling where each element or member of the population has an equal chance of being included in the sample. The most commonly used is the **lottery method**.

b. Systematic sampling - is another type of random sampling, which is known as **interval sampling.** This method considers an interval in selecting a sample from a given population. Using the formula:

nN k =

where: **k** = interval size, **N** = population size and **n**= sample size

c. Stratified sampling - is a random sampling method that divides a population into different homogenous subgroups called **strata**.

Two types of stratified sampling

c.1. Simple stratified sampling - is used when the population is divided into strata with common characteristic/s and if we decide to get an equal number of samples from each stratum.

c.2. Proportional stratified sampling -is used when the sample size is proportional to the number of members of the stratum. This means that the smaller the number of stratum members, the smaller the stratum's sample size will be.

d. Cluster sampling - usually used on a geographical basis and is sometimes called area sampling. It requires a complete list of clusters that represent the sampling frame.

e. Multi-stage sampling - it involves two or more stages in selecting the samples from a given population.

SAMPLING DISTRIBUTION OF THE SAMPLE MEANS – is frequency distribution of the sample means taken from a population.

Example 1

A population consists of five values (Php2, Php3, Php4, Php5, Php6). A sample of size 2 is to be taken from this population.

a. How many samples are possible? List them and compute the mean of each sample.

b. Construct the sampling distribution of the sample means.
c. Construct the histogram of the sampling distribution of the sample means.

Solution

1. Since the size of the population is 5, we have N = 5. We shall draw a sample of size 2 from this population, so n = 2. Thus, the number of possible samples of size 2 can be drawn from this population is computed as follows:

$$C(n,r) = \frac{n!}{(n-r)!r!}$$

$$C(5,2) = \frac{5!}{(5-2)!2!}$$

$$= 10$$

The number of all possible samples of size 2 is 10. The table shows the list of all possible samples with their corresponding means.

Possible samples of size 2	Mean
2 , 3 (2+3)/2	2.5
2 , 4	3.0
2 , 5	3.5
2 , 6	4.0
3 , 4	3.5
3 , 5	4.0
3 , 6	4.5
4 , 5	4.5
4 , 6	5.0
5 , 6	5.5

Observe that the means of the samples vary from sample to sample. The mean of the population μ=4, while the means of the samples may be less than, greater than, or equal to 4.

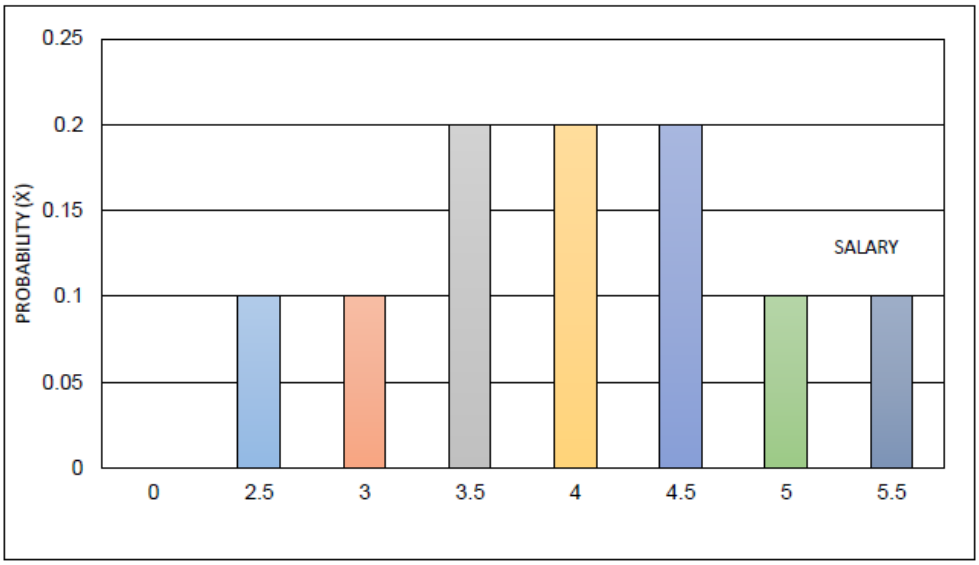
1. We now construct the frequency distribution of the sample means.

Mean	Frequency
2.5	1
3.0	1
3.5	2
4.0	2
4.5	2
5.0	1
5.5	1
Total	10

Next, we construct the probability distribution of the sample means. This is the sampling distribution of the sample means.

Mean $\bar{x}$	Probability P($\bar{x}$)
2.5	1/10
3.0	1/10
3.5	2/10 or 1/5
4.0	2/10 or 1/5
4.5	2/10 or 1/5
5.0	1/10
5.5	1/10

2. The histogram of the sampling distribution of the sample means is constructed by making a bar graph where the sample means are plotted on the horizontal axis and the corresponding probabilities are shown in the vertical axis.



Example 2

The following table gives the monthly salaries (in thousands of pesos) of six officers in a government office. Suppose that random samples of size 4 are taken from this population of six officers.

Officer	Salary
A	8
B	12
C	16
D	20
E	24
F	28

1. How many samples are possible? List them and compute the mean of each sample.
2. Construct the sampling distribution of the sample means.
3. Construct the histogram of the sampling distribution of the sample means.

Solution

Since the size of the population is 6, we have N = 6. We shall draw a sample of size 4. Thus, the number of possible samples of size 4 that can be drawn from this population is computed as follows.

$$C(n,r) = \frac{n!}{(n-r)!r!}$$

$$C(6,4) = \frac{6!}{(6-4)!4!}$$

= 15

The number of all possible samples of size 4 is 15. The table shows the list of all possible samples with their corresponding means.

Sample	Salaries	Mean
A, B, C, D	8, 12, 16, 20,	14
A, B, C, E	8, 12, 16, 24	15
A, B, C, F	8, 12, 16, 28	16
A, B, D, E	8, 12, 20, 24	16
A, B, D, F	8, 12, 20, 28	17
A, B, E, F	8, 12, 24, 28	18
A, C, D, E	8, 16, 20, 24	17
A, C, D, F	8, 16, 20, 28	18
A, C, E, F	8, 16, 24, 28	19
A, D, E, F	8, 20, 24, 28	20
B, C, D, E	12, 16, 20, 24	18
B, C, D, F	12, 16, 20, 28	19
B, C, E, F	12, 16, 24, 28	20
B, D, E, F	12, 20, 24, 28	21
C, D, E, F	16, 20, 24, 28	22

Observe that the means of the samples vary from sample to sample. The mean of the population $\mu = 18$, while the mean of the samples may be less than, greater than, or equal to 18.

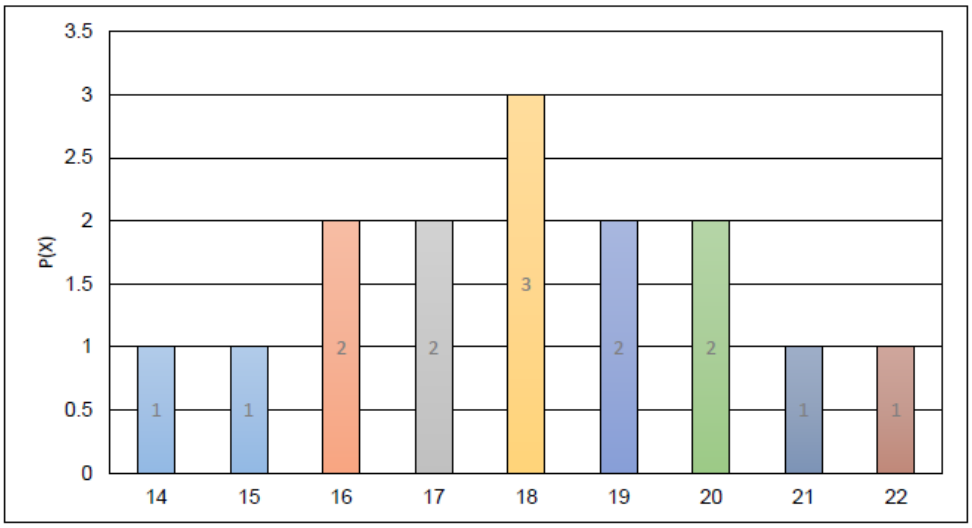
1. We now construct the frequency distribution of the sample means.

Mean($\bar{x}$)	Frequency (f)
14	1
15	1
16	2
17	2
18	3
19	2
20	2
21	1
22	1
Total	15

Next, we construct the probability distribution of the sample means. This is the sampling distribution of the sample means.

Mean $\bar{x}$	Probability P($\bar{x}$)
14	1/15
15	1/15
16	2/15
17	2/15
18	1/15
19	2/15
20	2/15
21	1/15
22	1/15

2. The histogram of the sampling distribution of the sample means is constructed by making a bar graph where the sample means are plotted on the horizontal axis and the corresponding probabilities are shown in the vertical axis.



PARAMETER AND STATISTIC

The main objective of conducting a survey is to estimate the value of some of the characteristics of a population. Let us consider one of the results of XYZ survey before the May 2016 presidential election. The actual percentage of all the voters represent the ***population parameter***, while the estimate of those percentage based from the sample is known as the ***sample statistic.***

The sampling method used in selecting the sample data strongly affects the quality of the sample statistic with regards to its representativeness and accuracy. The table below shows a list of the common symbols used for parameters and statistic:

Parameter	Statistic
Population mean (μ)	Sample mean ($\overline{X}$)
Population standard deviation (σ)	Sample standard deviation (s)
Population variance (σ^2)	Sample variance (s^2)
Population proportion (P)	Sample proportion ($\overline{p}$)

Examples

Identify the population parameter and sample statistic for each study.

A recent survey of 540 senior high school students in FTLSHS for the S.Y. 2019-2020 found that 90% of the students could be classified are good in Mathematics.

Population Parameter: All senior high school students in FTLSHS for the S.Y. 2019-2020

Sample Statistic: Collection of 540 senior high school students in FTLSHS for the S.Y. 2019-2020 or the 90% of all senior high school students in FTLSHS for the S.Y. 2019-2020

The average weight of every seventh person entering the Ayala mall within 3-hour period was 168 pounds.

Population Parameter: All the people entering the Ayala mall within the assigned 3-hour period

Sample Statistic: every seventh person entering the Ayala mall within 3-hour Period

What's More

SOLVE ME PLEASE!

Directions: Solve the following given.

The random sample size (n)	Finite population (N) consisting of	Solution	Final answer
1. n = 2	3, 4, 5		
2. n = 3	1, 2, 3, 4, 5		
3. n = 5	6, 7, 8, 9, 10, 11, 12		
4. n = 7	4, 6, 8, 10, 12, 14, 16, 18, 20		
5. n = 9	1, 2, 3, 4, 5, 6, 7, 8, 9, 10		
6. n = 5	Odd numbers between 9 – 21 inclusive		

What I Can Do

ANALYZE ME PLEASE!

Directions: Answer the following problems.

Random samples of size n = 2 are drawn from a finite population consisting of the number 5, 6, 7, 8, 9.

a. How many possible samples are there?
b. List all the possible samples and the corresponding mean for each sample.
c. Construct the sampling distribution of the sample means.
d. Construct the histogram for the sampling distribution of the sample means. Describe the shape of the histogram.

MODULE 6

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Estimation of Parameters. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.The module is divided into two lessons, namely:

Lesson 6 –Sampling and Sampling Distribution

After going through this module, you are expected to:

a. finds the mean and variance of the sampling distribution of the sample mean.
b. defines the sampling distribution of the sample mean for normal population when the variance is: (a) known; (b) unknown

Lesson 6 Sampling and Sampling Distribution

Statistician do not just describe the variations of the individual data values about the mean of the population. They are also interested to know how the means of the samples of the same size taken from the same population vary about the population mean. In this lesson, you will learn how to describe the sampling distribution of the sample means by computing its means and variance. You will also make a general conclusion regarding the mean and variance and shape of the sampling distribution and the sample means. There are many different possible samples of the same size that can be drawn from a given population. A statistic such as mean can be computed for each of the sample drawn.

What is It

If all possible random samples of size n taken **with replacement (independent)** from a population with a mean μ and variance σ^2, then the mean $(\mu_{\bar{x}})$, variance $(\sigma^2\bar{x})$ and standard deviation $(\sigma\bar{x})$ of the sampling distribution of the sample mean are:

$$\mu\bar{x} = \mu \text{ (mean)}, \ \sigma^2\,\bar{x} = \frac{\sigma^2}{n} \text{ (Variance)}$$

$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$(Standard deviation or standard error)

If all possible samples of size n are taken **without replacement (dependent**) from a finite population of size N with a mean μ and variance σ^2, then mean $(\mu_{\bar{x}})$, variance $(\sigma^2\bar{x})$ of the sampling distribution of the sample mean are:

$$\mu\bar{x} = \mu \text{ (mean)}, \ \sigma^2\bar{x} = \sigma^2 n\ (N-nN-1) \text{ (Variance)}$$

$\sigma\bar{x}=\sigma^2 n\ \sqrt{N-nN-1}$ (standard deviation or standard error)

Note: The factor $N-nN-1$ is called correction factor for finite population. It will be close to 1 and can be safely ignored when n is small compared to N.
Note that as we increase the sample size and the variance of the sample mean decreases.

THEOREM
If random samples of size n are taken from a population with a mean μ and standard deviation σ, then the sampling distribution of the sample mean $\bar{x}$ approaches normal distribution with mean $\mu\bar{x}$, and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, thus can be standardized as : $Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$

As the n increase, the sampling distribution of the sample mean gets nearer and nearer to the normal distribution .

- If σ in unknown, compute the sample standard deviation **s** then use it to replace σ in the formula if n ≥30.
- Even if n<30., the formula can still be used provided that the population is approximately normal and the population standard deviation σ is known.

Example 1 . Suppose a jar contains number 1,3 and 5. Show that $\mu_{\bar{x}}=\mu$ and

$$\sigma^2\bar{x} = \frac{\sigma^2}{n}.$$

Solution: The probability distribution is :

x	1	3	5
$f(x)$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
$xf(x)$	$\frac{1}{3}$	$\frac{3}{3}$	$\frac{5}{3}$
x^2	1	9	25
$x^2f(x)$	1/3	9/3	25/3

Since the distribution is uniform, that is , the observations have the same probabilities, the mean (μ) and sample variance (σ^2)can be easily computed as

$$\mu=1(\tfrac{1}{3}) +3(\tfrac{1}{3}) + 5 (\tfrac{1}{3}) = \tfrac{9}{3} = \mathbf{3}$$

$$\sigma^2 = \tfrac{1}{3}(1-3)^2+\tfrac{1}{3}(3-3)^2 +\tfrac{1}{3}(5-3)^2+\tfrac{1}{3} =\tfrac{4}{3} +0+\tfrac{4}{3} =\mathbf{\tfrac{8}{3}}$$

if we take two numbers in succession with replacement then, the possible 2 number samples are: (1,1) , (3,3), (5,5), (1,3), (3,1) (1,5), (5,1), (3,5) and (5,3). The average or mean of each pair, in order are 1,3,5,2,2,3,3, 4, and 4 respectively. IF we denote the means as random variable $\bar{x}$, then:

The probability distribution of $\bar{x}$ becomes:

x	1	2	3	4	5
$f(x)$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{2}{9}$	$\frac{1}{9}$
$xf(x)$	$\frac{1}{9}$	$\frac{4}{9}$	$\frac{9}{9}$	$\frac{8}{9}$	$\frac{5}{9}$
x^2	1	4	9	16	25
$x^2f(x)$	$\frac{1}{9}$	$\frac{8}{9}$	$\frac{27}{9}$	$\frac{32}{9}$	$\frac{25}{9}$

The mean of the sampling distribution (using the formula for mean of random variable is :

$$\mu_{\bar{x}} = \sum(\bar{x}\, f(\bar{x}))$$

$$= 1\left(\tfrac{1}{9}\right) + 2\left(\tfrac{2}{9}\right) + 3\left(\tfrac{3}{9}\right) + 4\left(\tfrac{2}{9}\right) + 5\left(\tfrac{1}{9}\right)$$

$$= \frac{1}{9} + \frac{4}{9} + \frac{9}{9} + \frac{8}{9} + \frac{5}{9}$$

$= \mathbf{\frac{27}{9} = 3}$, but $\mu = 3$, therefore , $\mu_{\bar{x}}$

The variance of the sampling distribution (using the formula for variance of random variable is:

$$\sigma^2\bar{x} = \mathrm{E}\,(x^2) - [E(x)]^2$$

$$= \sum(\overline{x})^2\, \mathrm{f}(\overline{x}) - \left[\sum \bar{x}\, f(\overline{x})\right]^2$$

$$= \left[1^2\left(\tfrac{1}{9}\right) + 2^2\left(\tfrac{2}{9}\right) + 3^2\left(\tfrac{3}{9}\right) + 4^2\left(\tfrac{2}{9}\right) + 5^2\left(\tfrac{1}{9}\right)\right] - (3)^2$$

$$= \frac{93}{9} - 9$$

$= \frac{4}{3}$, but $\sigma^2 = \frac{\left(\frac{8}{3}\right)}{2} = \mathbf{\frac{4}{3}}$

$\sigma^2\bar{x} = \frac{\sigma^2}{n}$, since the sample size **n** is 2.

Therefore:

$\boldsymbol{\sigma^2\bar{x} = \frac{\sigma^2}{n}}$

Example 2.

If a random sample of size 16 is taken with replacement from the population 1,1,1,2,4,5,5, and 6 . what is the mean , variance and standard deviation of the sampling distribution of the sample mean.

Answer: We can write the probability distribution of our population as:

X	1	.2	4	5	6
f(x)	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$

The mean of the distribution is:

$$\mu = 1\left(\tfrac{3}{8}\right) + 2\left(\tfrac{1}{8}\right) + 4\left(\tfrac{1}{8}\right) + 5\left(\tfrac{2}{8}\right) + 6\left(\tfrac{1}{8}\right)$$

$$= \frac{3}{8} + \frac{2}{8} + \frac{4}{8} + \frac{10}{8} + \frac{6}{8}$$

$= \mathbf{\frac{25}{8}}$ **or 3.13**

And the variance is:

$$\sigma^2 = 1^2\left(\tfrac{3}{8}\right) + 2^2\left(\tfrac{1}{8}\right) + 4^2\left(\tfrac{1}{8}\right) + 5^2\left(\tfrac{2}{8}\right) + 6^2\left(\tfrac{1}{8}\right) - \left(\tfrac{25}{8}\right)^2$$

$$= \frac{3}{8} + \frac{4}{8} + \frac{16}{8} + \frac{50}{8} + \frac{36}{8} - \frac{625}{64}$$

$= 13.625 - \frac{625}{64} = \mathbf{3.86}$

And the variance is:

$$\sigma^2 = 1^2(\tfrac{3}{8}) + 2^2(\tfrac{1}{8}) + 4^2(\tfrac{1}{8}) + 5^2(\tfrac{2}{8}) + 6^2(\tfrac{1}{8}) - (\tfrac{25}{8})^2$$

$$= \frac{3}{8} + \frac{4}{8} + \frac{16}{8} + \frac{50}{8} + \frac{36}{8} - \frac{625}{64}$$

$$= 13.625 - \frac{625}{64} = \mathbf{3.86}$$

Therefore, the mean of the sampling distribution ($\mu_{\bar{x}}$) of the sample mean is 3.13 (the same as μ) , while the variance of the sampling distribution ($\sigma_{\bar{x}}$) is $\frac{3.86}{16}$ = 0.24(that is σ^2/n). In addition, the standard deviation $\sigma_{\bar{x}}$ is $\sqrt{0.24}$ = 0.49 **(positive square root of the variance)**

Example 3.

A school has 5,000 students with a mean weight of 80 lbs., with a standard deviation of 20 lbs. If you draw a random sample of 50 students, what is the mean, variance and standard deviation (standard error) of the sampling distribution of the sample mean?

Answer: We know from the given that the μ = 80 lbs, therefore $\sigma_{\bar{x}}$ = 80 lbs. Since the sample (n=50) is taken from a finite population (N=5000) whose σ = 20, the standard error ($\sigma_{\bar{x}}$) can be computed as:

$$\sigma_{\bar{x}} = \frac{\sigma}{n}\sqrt{\frac{N-n}{N-1}} \qquad = \frac{20}{50}\sqrt{\frac{5000-50}{5000-1}}$$

$=2.828\sqrt{\frac{4950}{4999}}$ = (2.828)(0.995) = $\sigma_{\bar{x}}$ =**2.814** which is the square of the standard error is :

= $\sigma^2\ \bar{x}$=2.814^2 , =$\sigma^2\ \bar{x}$= 7.9185 or 7.919

What's More

A. Determine the mean($\mu_{\bar{x}}$) , variance($\sigma^2\bar{x}$) and standard deviation ($\sigma_{\bar{x}}$) of each.

(1-3) A random sample of 20 independent observations is taken from a population with μ=23.8 and σ=5.

1. $\mu_{\bar{x}}$ =_____
2. $\sigma^2\bar{x}$=_____
3. $\sigma_{\bar{x}}$=_____

(4-6). A random sample of size 30 independent observations is taken from a population with μ=48 and σ=6.5.

4. $\mu_{\bar{x}}$ =_____
5. $\sigma^2\bar{x}$=_____
6. $\sigma_{\bar{x}}$=_____

(7-9). A random sample of size 120 is taken with replacement from a population with independent μ=120 and σ=28.

7. $\mu_{\bar{x}}$ =_____
8. $\sigma^2\bar{x}$=_____

9. $\sigma_{\bar{x}}$=_____

What I Can Do

A. Compute the mean, variance and standard deviation of the sampling distribution taken from the following populations.

10-12. A community has 1500 people with mean age of 42 and variance of 16. If you draw a random sample of 30 people, what are the mean, variance And standard error of sampling distribution of their ages?

13-15. What is the mean, variance, and standard error of the sample mean when 60 students are taken from a population of 2000 with a mean score of 75 and standard deviation of 5?

MODULE 7

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Statistics and Probability. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using

This module targets the following learning competencies:

1. Illustrate the Central Limit Theorem (M11/12SP-IIIe-2).
2. Define the sampling distribution of the sample mean using the Central Limit Theorem (M11/12SP-III-3).

After going through this module, you are expected to:

- state and explain the Central Limit Theorem;
- identify the assumptions and conditions in using the Central Limit Theorem; and
- describe the sampling distribution of the sample means using the Central Limit Theorem by finding the mean and the standard error of the mean.

Lesson 7 The Central Limit Theorem

In the previous module, you have described the sampling distribution of the sample means by computing its mean and variance. This time, you will continue to study sampling distribution along with an important concept used in the field of Statistics.

What is It

A. The Central Limit Theorem

In the previous module, you have learned to use parameters in describing a population. For a standard normal distribution, the two parameters are the mean and standard deviation (or you may use the variance). You do not get the mean and then forget the standard deviation or the variance. These two go hand in hand. From there, you may be able to conclude about a population.

As mentioned in the previous part of this module, it is difficult and almost impossible to collect data from the entire population. So, what the researchers are doing is to obtain data from a sample, and then do the study. The result is then treated as an estimate of the population mean. If this is only an estimate of the population mean, then it is important to ensure that this is a good estimate. This is achieved when the standard deviation of the sampling distribution of the sample means (which is also known as standard error of the mean) is small or close to zero. The standard error of the mean determines how accurate the sample mean as an estimate of the population mean is. This value depends on the sample size. Remember that the formula to get the standard error of the mean ($\sigma_{\bar{x}}$) is determined by dividing the population standard deviation (σ) by the square root of the sample size (n). Meaning, to get a good estimate of the population mean, you have to increase the sample size. This is now where the Central Limit Theorem enters in.

The Central Limit Theorem (CLT) states that the sampling distribution of the sample means moves closer to a normal distribution as the sample size increases regardless of the shape of the population distribution. In other words, the distribution is approximately normally distributed whenever the sample size is sufficiently large irrespective of whether the distribution is normal, left-skewed, right-skewed, or uniform. This theorem also tells us that the mean of the sampling distribution of the sample means is always equal to the population mean.

B. Assumptions and Conditions for Using the CLT

There are assumptions and conditions that you have to consider before using the Central Limit Theorem.

1. The sample must be selected randomly. This is to secure that the data gathered are unbiased data from a population.
2. The variables must be independent from each other. Meaning, the value of one observation to another observation does not affect each other. Normally, when the sample is randomly selected, it is assumed that independence is already met.
3. The sample size should not be more than 10% of the population when the sample is drawn with no replacement. Taking away each item in the observations changes the

population. When you sample only 10% or less of the population, eliminating each observation does not change the population that much (Khan Academy).

4. The sample size must be large enough. How large? For many Statisticians, the minimum sample size of 30 is assumed to be sufficiently large. They believe that when the mean approaches a normal distribution. But of course, this is just a guideline and not a general rule because situation varies. If it is stated that the population is already normally distributed, you may use small sample size. However, when the population is strongly skewed, then you will need adequately large sample size. The basic concept is this: the more the distribution varies from being normal, the larger the sample size is required.

B. Defining the Sampling Distribution of the Sample Means Using the Central Limit Theorem

After knowing the Central Limit Theorem's definition, assumptions, and conditions, you may now describe the sampling distribution of the sample means by computing the mean and the standard error of the mean.

Example 1: The mean (μ) and standard deviation (σ) of population distribution are 68 and 3, respectively. Find the mean and standard deviation of the sampling distribution when a random sample of size 25 is drawn from the population. Assume that the population is finite.
Solution:

Steps	Solution
1. Enumerate the given.	μ = 68, σ = 3, n = 25
2. Find the mean of the sampling distribution.	Recall that $\mu_{\bar{x}} = \mu$ based on the CLT. So, the mean of the sampling distribution is automatically 68.
3. Find the standard deviation of the sampling distribution (also known as the standard error of the mean).	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{25}} = \frac{3}{5} = 0.6$

Example 2: Samples of size 100 are selected randomly from a population with a mean 28.5 and a standard deviation of 1.5. Compute the mean and standard error of the sampling distribution of the means given that the population is finite.
Solution:

Steps	Solution
1. Enumerate the given.	μ = 28.5, σ = 1.5, n = 100
2. Find the mean of the sampling distribution.	Recall that $\mu_{\bar{x}} = \mu$ based on the CLT. So, the mean of the sampling distribution is automatically 28.5.
3. Find the standard deviation of the sampling distribution (also known as the standard error of the mean).	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{100}} = \frac{1.5}{10} = 0.15$

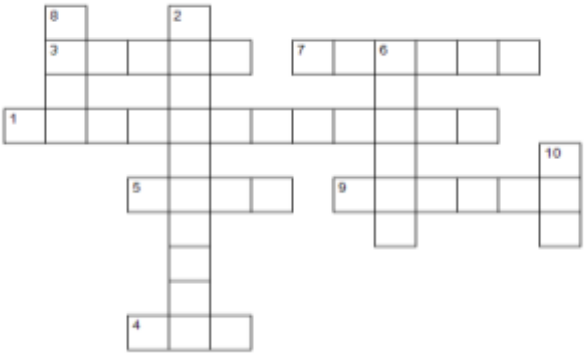
The Central Limit Theorem gives us confidence that whatever the shape of the distribution is, it will approach a normal distribution as the sample size increases.

In the next module, you will solve more real-life problems involving this theorem. Exciting, right?

What's More

Crossword Puzzle

Directions: Complete the puzzle by answering the given clues.



Across

1. this condition shows that the value of one observation to another observation does not affect each other
3. the relationship between the population mean and the mean of the sampling distribution of the sample means
4. the sample size should not exceed the _____ percent of the population when the sample is drawn without replacement
5. The standard error of the mean must be close to _____ to have a good estimate of the mean.
7. the more the distribution varies from being normal, the _____the sample size is required
9. the approximate shape of the sampling distribution of the sample means when the size of the random samples gets larger

Down

2. the standard error of the mean depends on this
6. the data collected must be from a _____ sample
8. a parameter for a standard normal distribution
10. abbreviation for Central Limit Theorem

What I Can Do

Instructions:

1. Do only the task that is assigned to your group.

   Group 1: Find the mean and standard deviation of the heights of all the students in your section.

   Group 2: Find the mean and standard deviation of the weights of all the students in your section.

   Group 3: Find the mean and standard deviation of the hours spent in sleeping of all the students in your section.

   Group 4: Find the mean and standard deviation of the waistline of all the students in your section.

   Group 5: Find the mean and standard deviation of the general average of all the students in your section.

2. After finding the mean and standard deviation, create a problem using the Central Limit Theorem for the solution.

3. Submit your work on a letter size bond paper before the end of the first quarter of this semester

RUBRIC:

	5	**3**	**1**	Score
Mathematical Concept	The solution has no error.	The solution has a minor error.	The solution has lots of errors.	
Presentation of the output	The submitted output shows neatness and orderliness.	The submitted output shows neatness and orderliness with minor flaws.	The submitted output is untidy and disorganized.	
Timeliness	The output is submitted on or before the deadline.	The output is submitted one or two days after the deadline.	The output is submitted later than two days after the deadline.	
TOTAL SCORE				

MODULE 8

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the random variable and probability distributions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module deals with an understanding of:

- Applied Problems Involving Sampling Distribution of Samples Means

After going through this module, you are expected to:

1. Solves problems involving sampling distributions of the sample means

Lesson 8	Applied Problems Involving Sampling Distribution of Sample Means

Your previous lesson on computing the mean and the variance of the sampling distribution of the means gives you an idea that the mean of the sampling distribution of the sample means μX is equal to the mean of the population. It is also shown to you from your lesson that the variance of the sampling distribution of the sample mean has two formulas to use, that is, if the population is finite and or infinite, as well as in finding the standard deviation since getting the standard deviation is just only taking the square root of the variance.

The knowledge taught to you in the previous lesson is pertinent in your succeeding lesson. But before we proceed to the next lesson, let me check your understanding.

What is It

Let us explore how the **Central Limit Theorem** is applied in solving problems involving sampling distribution.

A population of size of N = 60, has a mean μ= 65 and a standard deviation is σ=5

1. What is the probability that their mean is between 64 and 67?

2. What is the probability that a random sample of size of 35 will have a mean of 66 or more?

Let's take the step-by-step solution:

- the population is finite because its size is given and therefore, we can use the normal distribution as CLT holds.
  - ✓ 1st convert the values to z scores using $z = \frac{\bar{X} - \mu}{\sigma_{\bar{X}}}$
  - ✓ $z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{64-65}{\frac{5}{\sqrt{60}}} = \frac{-1}{0.645} = -1.55$

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{67-65}{\frac{5}{\sqrt{60}}} = \frac{2}{0.645} = 3.10$$

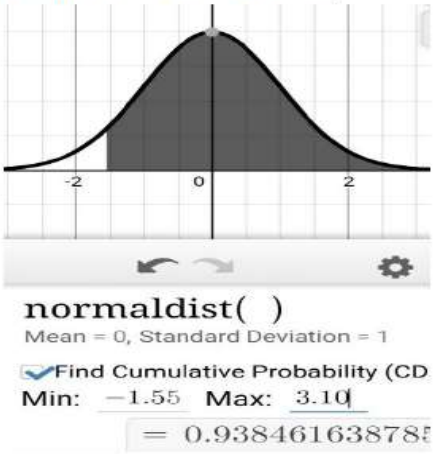
since the area below z =-1.55 is 0.0606, while the area below z =3.10 is 0.9990, then the area between z =-1.55 and z = 3.10, that is

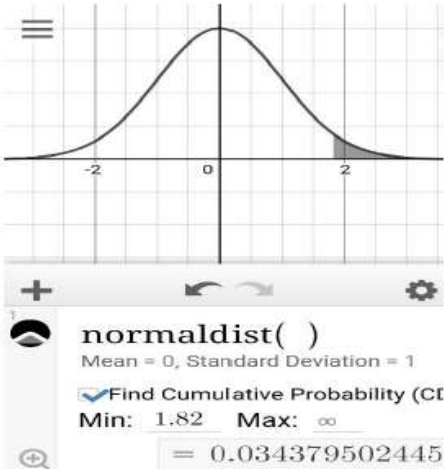
P(-1.55 < z < 3.10)

= 0.9990 – 0.606

= 0 9384 or

= 93.84%





What is the probability that a random sample of size of 35 will have a mean of 66 or more?

$P(X \geq 66 = P(z \geq 1.82)$

$= P(z \geq 0) - P(0 \leq z \leq 1.82)$

= 0.5000 – 0.4649

= 0.034379

or 3.4%

What's More

A. Compute the z value for each; assume tat each population is normally distributed.

(20 = 2 points each)

	1.	2.	3.
μ	100	140	62
σ	2	14	6
$\bar{X}$	100.5	145	59
n	80	12	30
$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$			

B.

	4.	5.
μ	46	245
σ²	9	20
$\bar{X}$	45.5	248
n	20	25
$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$		

What I Can Do

Try to do the following activities for to deepen your understanding on the Central Limit Theorem.

Assume that the height of adult women was normally distributed with a mean of 63 in and standard deviation of 2.5 in.

1. If 36 women are randomly selected, what is the probability that the mean height is less than 62 in?
2. If 70 women are taken as samples, what is the probability that their mean height is greater than 62.5 in?

MODULE 9

What I Need to Know

This module was designed, and This module was designed and written with you in mind. It is here to help you learn about T-Distribution and Percentile. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

The module is divided into 2 lessons, namely:

- Lesson 1 – Understanding the T-Distribution
- Lesson 2 –Percentile using T-Table

After going through this module, you are expected to:

1. Illustrate the T-Distribution
2. Identifies Percentiles using the T-Table

Lesson 9 Understanding the T-Distribution

There are situations when sample values are not large enough for the central Limit Theorem to be applied. Can we still obtain an interval estimate of the population mean? Are assumptions can be met? Those questions and other pertinent procedures were discussed in this lesson. To find out if you are ready to learn this new lesson, do the following activities.

What is It

N < 30

If the sample size is less than 30 (n<30), it is considered small, thus, even if the variance of the population is given, the formula for standardizing the sampling distribution of the sample mean cannot be used. For this small sample, the normality of the distribution sample mean cannot be guaranteed, thus, the z-table cannot be used.

The T-Distribution

Since the *z*-table cannot be used for small sample size, another type of distribution is used. This special case is called t-distribution, formulated in 1908 by an Irish brewing employee named W.S. Gosset.

The **t-distribution**, like the *z*-distribution, is a bell-shaped and symmetrical. But as compared to the *z*-distribution, the t-distribution is more variable since its value depends on the fluctuations of mean and variance from sample to sample.

Theorem

If $\bar{x}$ and s are the mean and standard deviation, respectively, of a random sample of size *n* taken from a normally distributed population with a mean μ, can be standardized as

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Where;

$\bar{x}$= sample mean

μ = population mean

s = sample standard deviation

n = sample size

a value of a random variable T following the t-distribution. Remember that the formula should be used when sample size (*n*) is less than 30.

The **t distribution** allows us to conduct statistical analyses on certain data sets that are not appropriate for analysis, using the normal distribution. The t-distribution can be used when your sample size (n) is less than 30 and the variance (σ) is unknown. Here is an example on how to find the t-value.

Example 1:

What is the t-value when μ=42, $\bar{x}$= 43, s=6 and n=25?

Steps	Solution
1. Analyze the problem and identify what kind of distribution is going to be used.	Since n=18 and it is less than 30, we can use the t-distribution
2. Write the formula	$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$
3. Evaluate the formula	$t = \frac{43 - 42}{\frac{6}{\sqrt{25}}}$
4. Subtract the numerator and find the square root of the sample size (n)	$t = \frac{1}{\frac{6}{5}}$

5. Divide the denominator	$t = \frac{1}{1.2}$
6. Simplify your answer	t = 0.8333 or 0.83

Example 2:

MATH Corporation manufactures light bulbs. The CEO claims that an average Acme light bulb lasts 400 days. A researcher randomly selects 20 bulbs for testing. The sampled bulbs last an average of 380 days, with a standard deviation of 50 days. Find the t-value of the given data.

Steps	Solution
1. Write the given	μ=400, $\bar{x}$=380, s=50, n=15
2. Analyze the problem and identify what kind of distribution is going to be used.	Since n=15 and it is less than 30, we can use the t-distribution
3. Write the formula	$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$
4. Evaluate the formula	$t = \frac{380 - 400}{\frac{50}{\sqrt{15}}}$
5. Subtract the numerator and find the square root of the sample size (n). If the square root is not a perfect square, write at least 4 decimal places then rounded off.	$t = \frac{-20}{\frac{50}{3.8730}}$
6. Divide the denominator	$t = \frac{-20}{12.9099}$
7. Simplify your answer	t = -1.5491 or -1.55

The T-Table

The t-table shows right-tail probabilities for selected t-distributions. You can use it to solve the following problems. Suppose you have a sample of size 10 and you want to find the 95th percentile of its corresponding t-distribution. You have n – 1= 9 degrees of freedom, so, using the t-table, you look at the row for df = 9. The 95th percentile is the number where 95% of the values lie below it and 5% lie above it, so you want the right-tail area to be 0.05. Move across the row, find the column for 0.05, and you get. This is the 95th percentile of the t-distribution with 9 degrees of freedom.

Now, if you increase the sample size to n = 20, the value of the 95th percentile decreases; look at the row for 20 – 1 = 19 degrees of freedom, and in the column for 0.05 (a right-tail probability of 0.05) you find degrees of freedom indicate a smaller standard

deviation and thus, the t-values are more concentrated about the mean, so you reach the 95th percentile with a value of t closer to 0.

***t* Table**	**t.80**	**t.90**	**t.95**	**t.98**	**t.99**	**t.998**	**t.999**
	***t* .90**	***t* .95**	***t* .975**	***t* .99**	***t* .995**	***t* .999**	***t* .9995**
Two-tails	0.20	0.10	0.050	0.02	0.010	0.002	0.001
one-tail	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
df							
1	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.690

28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
z	1.282	1.645	1.960	2.326	2.576	3.090	3.291

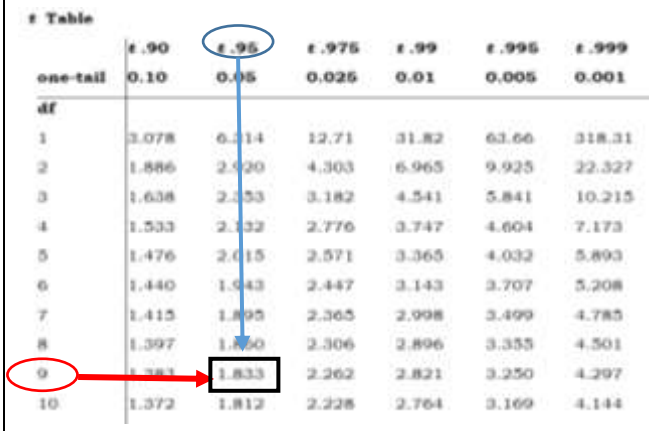
Note:

The formula for Degree of freedom (df) is n -1, Where n is the sample size.

$$df = n - 1$$

Example 1:

Mr. Sotto conducts a survey to 25 people for the effectiveness of their new medicine. He wants to know what is the 95th percentile of his survey. Find the t-value.

Steps	
Compute for the degree of freedom	df = n – 1 = 25 – 1 **df = 24**
Change the percentile into a percentage to decimal	95th = 95% = .95
Subtract the decimal to 1. To know what the right tail area is	α=1 - 0.95 α =**0.05**
Referring to the table. Look for the column of 0.05 and the row of the df.	
Write your answer	t-value = 1.833
Steps	
Compute for the degree of freedom	df = n – 1 = 25 – 1 **df = 24**
Change the percentile into a percentage to decimal	95th = 95% = .95
Subtract the decimal to 1. To know what the right tail area is	α=1 - 0.95 α =**0.05**

t Table

	t .90	t .95	t .975	t .99	t .995	t .999
one-tail	0.10	0.05	0.025	0.01	0.005	0.001
df						
1	3.078	6.314	12.71	31.82	63.66	318.31
2	1.886	2.920	4.303	6.965	9.925	22.327
3	1.638	2.353	3.182	4.541	5.841	10.215
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.893
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144

Referring to the table. Look for the column of 0.05 and the row of the df.	t Table t .90 / t .95 / t .975 / t .99 / t .995 / t .999 one-tail 0.10 / 0.05 / 0.025 / 0.01 / 0.005 / 0.001 df 1 3.078 6.314 12.71 31.82 63.66 318.31 2 1.886 2.920 4.303 6.965 9.925 22.327 3 1.638 2.353 3.182 4.541 5.841 10.215 4 1.533 2.132 2.776 3.747 4.604 7.173 5 1.476 2.015 2.571 3.365 4.032 5.893 6 1.440 1.943 2.447 3.143 3.707 5.208 7 1.415 1.895 2.365 2.998 3.499 4.785 8 1.397 1.860 2.306 2.896 3.355 4.501 9 1.383 1.833 2.262 2.821 3.250 4.297 10 1.372 1.812 2.228 2.764 3.169 4.144
Write your answer	t-value = 1.833

Example 2: What is the t-value when n=15 at 90th percentile (one tail)
df =n-1 = 15-1 = **14** and 90th percentile is α=0.10.
Then referring to the table, **t = 1.345.**

What's More

A. Compute the t-value for each number below. (2 Points Each)

1. μ=36, $\bar{x}$=30 , s= 5 n= 20
2. μ=536 , $\bar{x}$=550 , s= 14 n= 9
3. μ=47.2, $\bar{x}$=48.9, s=5 n= 12
4. μ=77 , $\bar{x}$=82 , s= 12 n= 22
5. μ=200 , $\bar{x}$= 189, s= 19 n= 16

Complete the table and find the t-value on the given problems. (5 points)

1. Suppose scores on a Math test is normally distributed, with a population mean of 100. Suppose 20 people are randomly selected and tested. The standard deviation in the sample group is 15. What is the probability that the average test score in the sample group will be at most 110?

a. What is/are given?
b. Write the formula to be used
c. Solution

What I Can Do

Answer the given questions.

1. The mean percentage score of 20 students answering the module is 89% with a standard deviation of 1.6. But the teacher claims that the percentage score is 92%. Find the t-value for this question.
2. Ms. Pelaez conducts a written evaluation to her 25 students. She wants to know the 95th percentile of her assessment. Find its t-value.

MODULE 10

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the Estimation of Parameters. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into two lessons, namely:

Lesson 10 – Length of Confidence Interval and Appropriate Sample Size

After going through this module, you are expected to:

1. Identify the length of a confidence interval;
2. computes for the length of the confidence interval;
3. computes for an appropriate sample size using the length of the interval;
4. solves problems involving sample size determination.

Lesson 10 Length of Confidence Interval and Appropriate Sample Size

Researchers cannot always study the entire population especially when the population especially when the population is too large. When the population is too large, it may not be possible to determine the population parameters such as the population variance and population mean, population variance, and population standard deviation. Having a slice of the population whose elements were randomly selected from the entire population is not enough. Aside from the sampling technique discussed in Unit 2., researchers need additional techniques to be able to make inferences about the entire population. These techniques are discussed in this unit- one will learn the best point estimates for the population mean and population proportion, how to compute these using point estimators, how to compute for the confidence interval estimate based on the appropriate form of estimator based on the appropriate form of estimator for the population mean, and how to solve problems involving confidence interval estimation of population proportion.

What is It

Length of Confidence Interval and Appropriate Sample Size

In statistics, a **confidence interval** (CI) is a type of estimate computed from the statistics of the observed data. This proposes a range of plausible values for an unknown parameter (for example, the mean). The interval has an associated confidence level that the true parameter is in the proposed range.

Interval estimate is sometimes called **Confidence Interval**, is a or interval with lower and upper limits used to estimate the population parameter. It is usually in the form a <θ <b,which tells that the estimated parameter θ is between two values (a and b) at a certain level of confidence.

Statistical inference is a making conclusion or generalization about the population based on the study of samples.

Point Estimate is a single value that estimates the population parameter, such as $\bar{x}$ as estimate for μ, or s as estimate for α.

More strictly speaking, the confidence level represents the frequency (i.e., the proportion) of possible confidence intervals that contain the true value of the unknown population parameter. In other words, if confidence intervals are constructed using a given confidence level from an infinite number of independent sample statistics, the proportion of those intervals that contain the true value of the parameter will be equal to the confidence level. For example, if the confidence level (CL) is 90% then in hypothetical indefinite data collection, in 90% of the samples the interval estimate will contain the population parameter.

When the population variance or standard deviation σ is known, or when $n \geq 30$(by Central limit Theorem), the formula below can be used an interval estimate of population mean μ at a certain degree of confidence (α):

$$\bar{x} - z_{\frac{a}{2}}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \bar{x} + z_{\frac{a}{2}}\left(\frac{\sigma}{\sqrt{n}}\right)$$

Where : $\bar{x}$ = sample mean ,

σ = population standard deviation

n= sample size

$z_{\frac{a}{2}}$ = z value that leaves an area of $\frac{a}{2}$

The values of $z_{\frac{a}{2}}$ are listed below with the usual confidence level used in estimating population mean.

Confidence level or (1-a)100%	90%	95%	97%	98%	99%
$z_{\frac{a}{2}}$	1.64	1.96	2.17	2.33	2.58

From the formula , $z_{\frac{\alpha}{2}}(\frac{\sigma}{\sqrt{n}})$ is called **margin of error.**

Definition
Margin of error refers to the maximum acceptable difference (determined by α) between the observed sample statistic (mean or proportion) and the true population parameter (mean or proportion).

The length of a confidence interval is the absolute difference between the upper confidence limit and the lower confidence limit. That is,

LCI=$|UCL - LCL|$=$|LCL - UCL|$ or LCI = UCL – LCL

Where : LCI = length of Confidence Interval

UCL = Upper Confidence Limit

LCL = Lower Confidence Limit

Example No.1 : Find the length of the following confidence interval

a. 0.357 <$\hat{p}$ < 0.603
b. 0.629 <$\hat{p}$ < 0.655

Solution:

a. LCI = UCL – LCL
=0.603 – 0.357
=**0.246**

b. LCI = UCL – LCL
=0.655 – 0.629
=**0.026**

- The formula for a confidence interval for population proportion p is

$$\hat{p}\text{-}z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}\ (1-\hat{p})}{n}} < \text{p} < \hat{p} + 1z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}\ (1-\hat{p})}{n}}$$

$$\text{LCI} = \left[\hat{p} + z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}\ (1-\hat{p})}{n}}\right] - \left[\hat{p} - z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}\ (1-\hat{p})}{n}}\right]$$

$$= \hat{p} + z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}\ (1-\hat{p})}{n}} + \hat{p} + z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}\ (1-\hat{p})}{n}}$$

$$\text{LCI} = 2z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{P}\ (1-\hat{p})}{n}}$$

The last equation above can be used to find the length of a confidence interval for population proportion.

Example 2. Find the length of the confidence interval given the following data

$\hat{p} = 0.25\ , n = 400, confidence\ level:\ 95\%$

Solution:

1. Find α in (1- α) 100% confidence level then find $z_{\frac{\alpha}{2}}$.

   (1- α) 100% = 95%

   1- α = 0.95

   α = 0.05

   $\frac{\alpha}{2} = \frac{0.05}{2}$ = 0.025

   0.5 – 0.025 = 0.475

   Hence, using the areas under the standard Normal Curve Table, $z_{\frac{\alpha}{2}}$=1.96

2. LCI = $2z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}\ (1-\hat{p})}{n}}$

   $= 2(1.96)\sqrt{\frac{0.25\ (0.75)}{400}}$

   $= 3.92\sqrt{\frac{0.1875}{400}}$

   =**0.0848 or 0.085**

The last equation above can be used to find the length of the confidence interval.

Example 3. Find the length of the confidence interval, given the following data.

s=6.17 , n=12, confidence level : 99%

Solution:

1. Find the degrees of freedom df.

   df= n−1

   = 12 −1

   =11

2. Find α in (1 −α) 100% confidence level then find $z_{\frac{\alpha}{2}}$.

   (1 − α) 100% = 99%

   1 − α = 0.99

   α = 0.01

   $\frac{\alpha}{2} = \frac{0.01}{2}$ = 0.005

   0.5 – 0.005 = 0.47

Hence, using the areas under the standard Normal Curve Table, $z_{\frac{\alpha}{2}}$=3.106.

3. LCI = $2t_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}}$

   $=2(3.106)\ (\frac{6.17}{\sqrt{12}})$

   $=(6.212)\ (\frac{6.17}{\sqrt{12}})$

   =**11.064 or 11.06**

Where : n=desired sample size

$\hat{p}$ = sample proportion

$z_{\frac{\alpha}{2}}$ =z value

E= margin error

Determining the Sample of Size for Estimating *p*

The following are the formulas:

A. When an estimate $\hat{p}$ is known:

$$n=\frac{\left[z_{\frac{\alpha}{2}}\right]^2 \hat{p}(1-\hat{p})}{E^2}$$

B. When an estimate $\hat{p}$ is unknown

$$n=\frac{0.25\left(z_{\frac{\alpha}{2}}\right)^2}{E^2}$$

Where : n=desired sample size

$\hat{p}$ = sample proportion

$z_{\frac{\alpha}{2}}$ =z value

E= margin error

0.25 is constant

If the computed sample size is not a whole number, it should be rounded up to the next whole number.

Example 4.

In previous study done by a student, it was found out that 28.5% of the student used twitter. This year, your statistics teacher wants you to conduct a study on the current percentage of Twitter users among the students in your school. How many students must you include in your study to be 95% confident so that the margin of error is no more than 3.5 percentage?

Solution:

a. *P*= 28.5 %
=0.285

q hat
1-0.285 = 0.715

b. E= 3.5%
=0.035

$(1-\alpha)$ 100% = 95%

$1-\alpha$ = 0.95

α = 0.05

$\frac{\alpha}{2} = \frac{0.05}{2}$ = 0.025

0.5 – 0.025 = **0.475** , hence $z_{\frac{\alpha}{2}}$=1.96.

c. $=\frac{\left[z_{\frac{\alpha}{2}}\right]^2 \hat{p}(1-\hat{p})}{E^2}$

$= \frac{[1.96]^2(0.285)\ (0.715)}{(0.035)^2}$

=**639.04 ≈ 640**

What's More

A. Find the length of the following confidence interval.

1. $0.325 < p < 0.575$
2. $0.137 < p < 0.563$
3. $0.338 < p < 0.562$
4. $0.301 < p < 0.751$
5. $0.245 < p < 0.467$

B. Find the length of the confidence interval given the following upper and lower limits.

6. Upper confidence limit= 0.673
   Lower confidence limit = 0.447
7. Upper confidence limit= 0.691
   Lower confidence limit = 0.409
8. Upper confidence limit= 0.765
   Lower confidence limit = 0.535
9. Upper confidence limit= 0.861
   Lower confidence limit = 0.619
10. Upper confidence limit= 0.852
    Lower confidence limit = 0.588

What I Can Do

C. Find the length of the confidence interval given the following data.

11. $\hat{P}$ =0.35,	n=400,	confidence level= 95%
12. $\hat{P}$ =0.45,	n=350,	confidence level= 95%
13. $\hat{P}$ =0.48,	n=410,	confidence level= 95%
14. $\hat{P}$ =0.51,	n=420,	confidence level= 95%
15. $\hat{P}$ =0.42,	n=300,	confidence level= 95%
16. σ=0.5,	n=36,	confidence level =95%
17. σ=0.4,	n=64,	confidence level =95%
18. σ=0.52 ,	n=40,	confidence level =95%
19. s = 5.25	n=14,	confidence level =95%
20. s = 6.05	n=12,	confidence level =95%

Quarter II

MODULE 1

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Statistics and Probability. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module consists of one lesson which contains sub lessons:

- Lesson 1 – Basic Concepts of Hypothesis Testing
    - 1.1. Null hypothesis
    - 1.2. Alternative hypothesis
    - 1.3. Level of Significance
    - 1.4. Rejection Region
    - 1.5. Types of Error in hypothesis Testing
    - 1.6. The Parameter tested to be tested given a real-life problem

After going through this module, you are expected to:

1. Illustrates: (a) null hypothesis; (b) alternative hypothesis; (c) level of significance; (d) rejection region; (e) types of errors in hypothesis testing.
1. Identifies the parameter to be tested given a real-life problem.

Lesson 1 Tests of Hypothesis

In daily life, we make tentative explanation of facts about a particular phenomenon by formulating hypothesis. This hypothesis may be correct or incorrect, depending on the available evidence that we can gather to support our hypothesis. We usually use a sample to gather information and evidence that we need to validate our hypothesis. The data that we gather from this sample become the basis of our decision whether we shall accept or reject our hypothesis regarding the entire population. The data obtained from this sample is analyzed with the use of appropriate statistical procedure to find out whether our hypothesis should be accepted. This process is called testing hypothesis. In this lesson 1 we shall discuss another aspect of inferential statistics: the testing of hypothesis. We shall learn how to conduct a test of hypothesis that will help us to arrive at the right decision.

What is It

A **statistical hypothesis** is a statement about the numerical value of a population parameter. It is a statement or tentative assertion which aims to explain facts about a certain phenomenon. A hypothesis needs to be resolved whether it is true or not. Thus, it must be subjected to statistical testing procedure known as **test of hypothesis or hypothesis testing**. If the hypothesis is **found to be true**, it is **accepted**: if **it is found false**, it is **rejected**.

There are two kinds of hypothesis: the null and alternative hypotheses.

1. A **null hypothesis**, denoted by ***Ho***, is a statement that there is no difference between a parameter and a specific value.
2. An **alternative hypothesis** denoted by ***Ha***, is a statement that there exists a difference between a parameter and a specific value. It is the opposite or negation of the null hypothesis.

In a **hypothesis test**, sample data is evaluated in order to arrive at a decision about some type of claim. If certain conditions about the sample are satisfied, then the claim can be evaluated for a population. In a hypothesis test, we: Evaluate the **null hypothesis**, typically denoted with H_0. The null is not rejected unless the hypothesis test shows otherwise. The null statement must always contain some form of equality (=, ≤ or ≥) Always write the **alternative hypothesis**, typically denoted with H_a or H_1, using less than, greater than, or not equals symbols, i.e., (≠, >, or <). If we reject the null hypothesis, then we can assume there is enough evidence to support the alternative hypothesis. Never state that a claim is proven true or false. Keep in mind the underlying fact that hypothesis testing is based on probability laws; therefore, we can talk only in terms of non-absolute certainties.

Since the null and alternative hypotheses are contradictory, you must examine evidence to decide if you have enough evidence to reject the null hypothesis or not. The evidence is in the form of sample data.

After you have determined which hypothesis the sample supports, you make a decision. There are two options for a **decision**. They are "reject H_0" if the sample information favors the alternative hypothesis or "do not reject H_0" or "decline to reject H_0" if the sample information is insufficient to reject the null hypothesis

Study the following examples:

Example 1

Claim: The average monthly income of Filipino families who belong to low-income bracket is Php8 000.

H_o: The average monthly income of Filipino families who belong to low income bracket is Php8 000 **(μ = 8 000).**

H_a: The average monthly income of Filipino families who belong to low income bracket is not equal to Php8 000 **(μ ≠ 8 000).**

Notice that the null hypothesis is expressed through the use of the "equal" symbol while the alternative hypothesis is expressed by the "not equal" symbol because the claim or conjecture does not specify any direction.

Example 2

H_0: No more than 30% of the registered voters in Santa Clara County voted in the primary election. **$\mu \leq 30\%$**

H_a: More than 30% of the registered voters in Santa Clara County voted in the primary election. **$\mu > 30\%$**

Example 3

We want to test if college students take less than five years to graduate from college, on the average. The null and alternative hypotheses are:

H_0: $\mu \geq 5$

H_a: $\mu < 5$

In formulating the hypotheses, we can use the following guidelines:

1. A null hypothesis is generally a statement of no change. Thus, a statement of equality or one which involves the equality is usually considered in the null hypothesis. Possible forms of the null hypothesis include (a) equality; (b) less than or equal; and (c) greater than or equal.
2. The statistical hypothesis is about a parameter or distribution of the population values. For example, the parameter in the statement is the average daily number of text messages that a Grade 11 student sends. Usually, the parameter is represented by a symbol, like for the population mean, we use μ. Hence, the null and alternative hypotheses could be stated using symbols as "Ho: μ = 100 against Ha: μ ≠ 100."
3. The null and alternative hypotheses are complementary and must not overlap. The usual pairs are as follow:

(a) Ho: Parameter = Value versus Ha: Parameter ≠ Value.
(b) Ho: Parameter = Value versus Ha: Parameter < Value.
(c) Ho: Parameter = Value versus Ha: Parameter > Value.
(d) Ho: Parameter ≤ Value versus Ha: Parameter > Value, and
(e) Ho: Parameter ≥ Value versus Ha: Parameter < Value.

Types of Tests

A statistical test may either be directional (one-tailed) or non-directional (two-tailed). We can determine whether a test is directional or nondirectional by looking at how alternative hypothesis is expressed.

Directional Test

A test of any statistical hypothesis where the alternative hypothesis is expressed using less than (<) or greater than(>) is called directional test or one-tailed test since the critical or rejection region lies entirely in one tail of the sampling distribution.

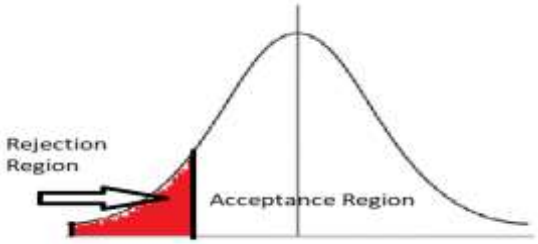
Study the following examples.

Example 4

Claim: The average weekly allowance of college students is less than Php 1 500.

H_0: The average weekly allowance of college students is equal to Php 1 500 (μ = 1 500).

H_a: The average weekly allowance of college students is less than Php 1 500 (μ < 1 500).



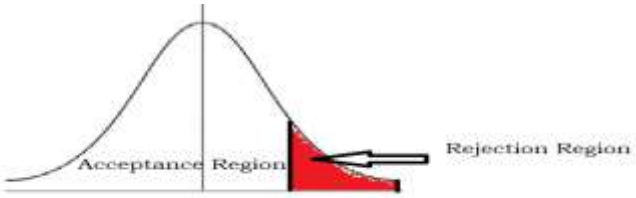
This is directional test or one-tailed test. More specifically, this is a left-tailed test because the "less than" symbol was used in expressing the alternative hypothesis Thus, the critical region or the rejection region lies entirely in the left tail of the sampling distribution.

Example 5

Claim: The average weekly allowance of college students is greater than Php 1 500.

H_0: The average weekly allowance of college students is equal to Php 1 500 (μ = 1 500).

H_a: The average weekly allowance of college students is less than Php 1 500 (μ > 1 500).



This is also directional test or one-tailed test. More specifically this is a right-tailed test because the “greater region than" symbol was used in expressing the alternative hypothesis. Thus, the critical region or the rejection region lies entirely at the right tail of the sampling distribution.

Nondirectional Test

A test of any statistical hypothesis where the alternative hypothesis is written with a not equal sign (≠) is called a nondirectional test or two-tailed test since there is no assertion made on the direction of the difference. The rejection region is split into two equal parts, one in each tail of the sampling distribution.

Example 6

Claim: The average weekly allowance of college students is Php 1 500.

H_0: The average weekly allowance of college students is equal to Php 1 500 (μ = 1 500).

H_a: The average weekly allowance of college students is less than Php 1 500 ($\mu \neq$ 1 500).

Observe that the alternative hypothesis is expressed, using the "not equal" (≠) symbol; the test is two-tailed.

Types of Error

In decision-making, we sometimes make a wrong decision. Likewise. when we test a hypothesis, there is a possibility that we shall also commit an error of accepting or rejecting the hypothesis. There are two types of errors: The Type I error and the Type Il error.

Type I error occurs when we reject the null hypothesis when it is true. It is also called **alpha error (α error).**

Type II error occurs when we accept the null hypothesis when it is false. It is also called **beta error (ß error).**

In hypothesis testing, four outcomes are possible: two of which lead to incorrect decisions. The four possible outcomes are described in the table below.

Decision	Fact	
	H_o is true	*H_o is false*
Accept H_o	*Correct decision*	*Type II error*
Reject H_o	*Type I error*	*Correct decision*

Level of Significance

The probability of committing Type I error is called the level of significance, It is denoted by the Greek letter α (alpha). Thus. the value of a tells us the probability of making an error in rejecting the null hypothesis when it is true. The choice for the value of the significance level is determined by the researcher. This depends on the risk or degree of confidence the researcher is willing to take in committing Type I error. The commonly used levels of significance are 0.05 and 0.01. The level of significance should be set before testing the hypothesis.

Example 7

A 0.01 level of significance means that the researcher is willing to take 1% error in making a decision. 't also implies that he is confident that he will make a right decision. Likewise, a 0.05 level of significance means that the researcher is willing to take error in making a decision. It also implies that he is 95% confident that he will make a right decision.

Steps in Testing the Hypothesis

Whenever we test hypotheses, we follow these steps.

Step 1: Identity the claim and formulate the null (H_O) and alternative (H_a) hypothesis

Step 2: Set the level of significance and determine whether the test is one-tailed or two-tailed by looking at how the alternative hypothesis is expressed, Decide on the test statistic to be used and find the critical value for the test. Draw or illustrate the rejection region.

Step 3: Compute the test value, using the test statistic or formula for the test.

Step 4: Make a decision whether to accept or reject the null hypothesis.

Step 5: Formulate a conclusion by answering the research question.

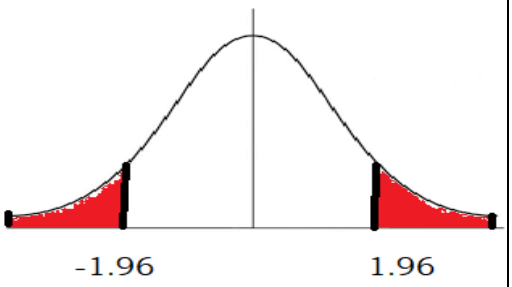
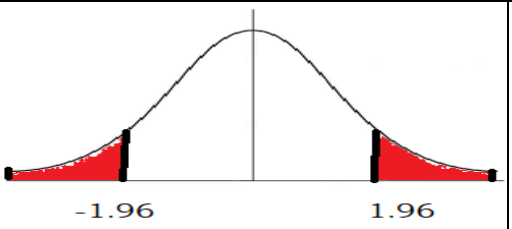
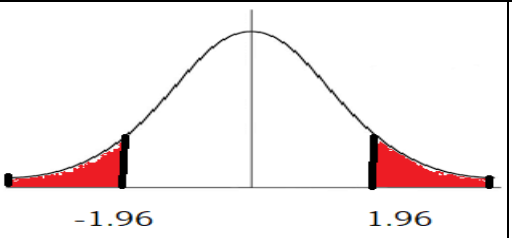
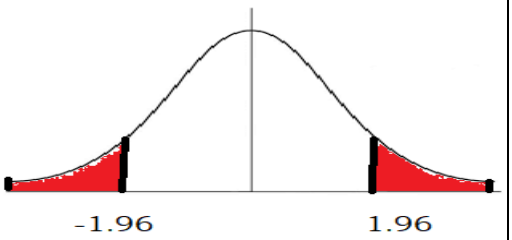
Accepting or Rejecting the Null Hypothesis

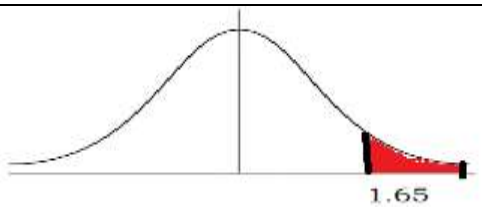
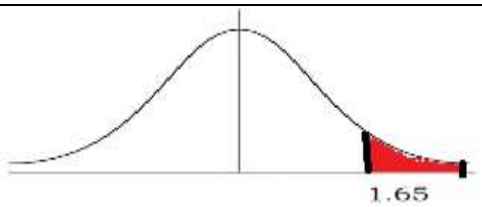
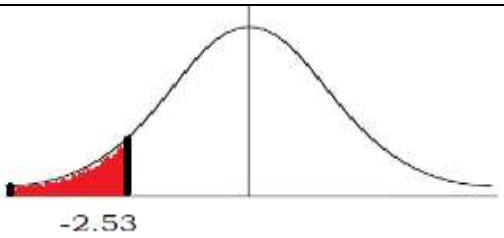
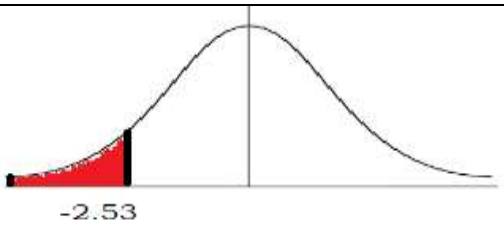
How do we decide on accepting or rejecting the null hypothesis? Follow these steps.

1. Determine the critical value. using appropriate statistical tables,
2. Draw the rejection region and the critical value.
3. If the test value or the computed value falls in the rejection region, then reject the null hypothesis; otherwise, accept the null hypothesis.

Study the following examples:

Example 8

Hypotheses	Rejection Region	Decision
H_0: μ = 50 H_a: μ ≠ 50 Test Value = 1.45 Critical Value = ±1.96		Accept H_0
H_0: μ = 50 H_a: μ ≠ 50 Test Value = -1.65 Critical Value = ±1.96		Accept H_0
H_0: μ = 50 H_a: μ ≠ 50 Test Value = 2.35 Critical Value = ±1.96		Reject H_0
H_0: μ = 50 H_a: μ ≠ 50 Test Value = -2.35 Critical Value = ±1.96		Reject H_0

H_0: μ ≤ 50 H_a: μ > 50 Test Value = 1.86 Critical Value = 1.65		Fail to Accept/Reject H_0
H_0: μ ≤ 50 H_a: μ > 50 Test Value = 1.34 Critical Value = 1.65		Accept H_0
H_0: μ ≥ 50 H_a: μ < 50 Test Value = -2.05 Critical Value = −2.53		Accept H_0
H_0: μ ≥ 50 H_a: μ < 50 Test Value = -2.88 Critical Value = −2.53		Fail to accept/Reject H_0

What's More

Identify Me Please!

DIRECTIONS: Identify whether the following is a **null hypothesis (H_o)** or an **alternative hypothesis (H_a)**.

Statement
1. The average age of grade eleven students is 17 years old.
2. The mean content of citric acid in a bottle of juice drinks is greater than 2 ml.
3. The average monthly salary of private school teachers is less than Php16 000.
4. The mean weight of newborn babies is 0.5kg.
5. The average IQ of grade eleven students is less than 108.
6. The mean starting salary for education graduates is at least Php 250 000 per year.
7. The mean number of years Americans work before retiring is 34.
8. The average height of mango tree in a farm is at least 15 meters.

9. The average score of grades eleven students in Filemon T. Lizan Senior High in Statistics and Probability during the Diagnostic Test is at most 45 out of 50-item test.
10. The mean number of cars a person owns in his/ her lifetime is not more than ten.

What I Can Do

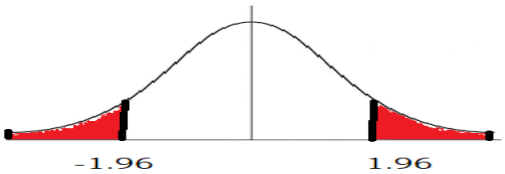
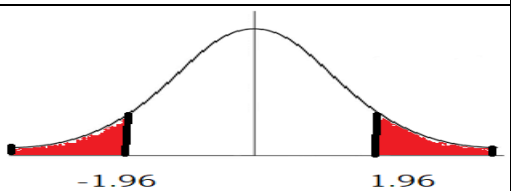
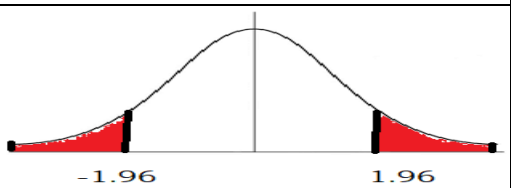
I Can Do These!

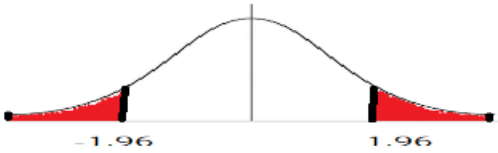
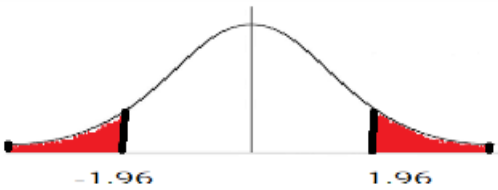
DIRECTIONS: Identify whether the test of hypothesis to be performed is ***one-tailed*** *or* ***two-tailed.***

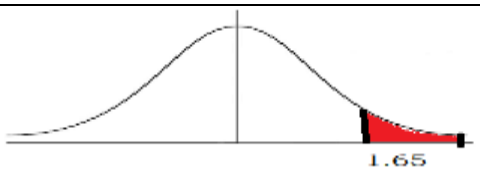
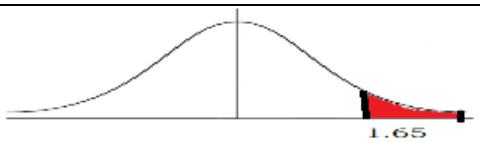
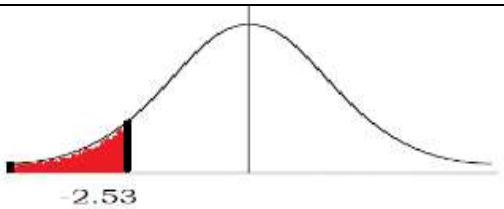
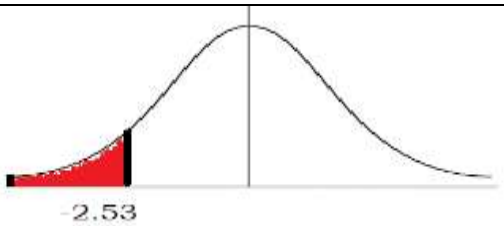
1. The average time to commute from home to school is 32.8 minutes.
2. The average number of vehicles passing through NLEX daily is less than 21 000.
3. The average daily number of customers in a convenient store is less than 1 025.
4. The mean content of citric acid in a bottle of juice drinks is greater than 2 ml.
5. The average typing speed of a secretary is 23.8 words per minute.
6. H_O: $\mu = 12$ H_a: $\mu \neq 12$
7. H_O: $\mu \leq 10$ H_a: $\mu > 10$
8. H_O: $\mu \geq 12$ H_a: $\mu < 12$
9. H_O: $\mu \leq 12$ H_a: $\mu > 12$
10. H_O: $\mu = 10$ H_a: $\mu \neq 10$

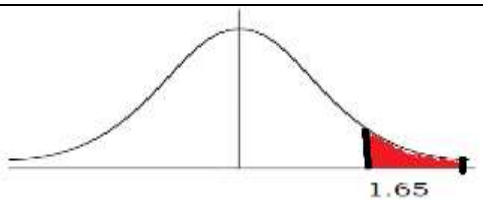
What's My Decision?

Directions: Decide whether the null hypothesis is to be accepted or rejected, given the test value and the critical value of test statistic.

Hypotheses	Rejection Region	Decision
1. H_0: $\mu = 150$ H_a: $\mu \neq 150$ Test Value = 2.35 Critical Value = ± 1.96		_____
2. H_0: $\mu = 150$ H_a: $\mu \neq 150$ Test Value = -1.34 Critical Value = ± 1.96		_____
3. H_0: $\mu = 150$ H_a: $\mu \neq 150$ Test Value = 1.97		_____

Critical Value = ±1.96		
4. H_0: μ = 150 H_a: μ ≠ 150 Test Value = -2.02 Critical Value = ±1.96		_____
5. H_0: μ = 150 H_a: μ ≠ 150 Test Value = -1.99 Critical Value = ±1.96		_____

6. H_0: μ ≤ 150 H_a: μ > 150 Test Value = 1.56 Critical Value = 1.65		_____
7. H_0: μ ≤ 150 H_a: μ > 150 Test Value = 1.28 Critical Value = 1.65		_____
8. H_0: μ ≥ 150 H_a: μ < 150 Test Value = -2.55 Critical Value = −2.53		_____
9. H_0: μ ≥ 150 H_a: μ < 150 Test Value = -3.01 Critical Value = −2.53		_____

10. H_0: μ ≤ 150 H_a: μ > 150 Test Value = 1.76 Critical Value = 1.65		_____

MODULE 2

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the random variable and probability distributions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is focus only in one topic:

Lesson 1 – Hypothesis Testing About a Population Mean When the Variance is known

Lesson 2 – Hypothesis Testing About a Population Mean When the Variance is Unknown

Learning Competencies:

- ✓ formulates the appropriate null and alternative hypotheses on a population mean. (M11/12SP-IVb-1)
- ✓ identifies the appropriate form of the test-statistic when: (a) the population variance is assumed to be known; (b) the population variance is assumed to be unknown; and (c) the Central Limit Theorem is to be used. (M11/12SP-IVb-2)

After going through this module, you are expected to:

- ✓ differentiate traditional approach from probability value approach of hypothesis testing
- ✓ determine whether a hypothesis test is non-directional or directional
- ✓ determine whether a directional test is left -tailed or right -tailed

Lesson 2 Hypothesis Testing About a Population Mean When the Variance is known

We make decision all the time, consciously or not. In studying statistic, making decisions based on observation or data that are considered random variable is an essential concept to learn. Such procedure making decision is called **hypothesis testing.**

One of the ultimate goals of every nation is to produce professional who will contribute to scientific knowledge through research. In research investigation, hypothesis testing is a vital procedure. It is deciding whether to accept and reject a statement or the assumption about some parameter in any research problem. From the results of the correct decision making, conclusions are drawn in which facts are generated, and thus can become a contribution to a body of knowledge in the fields of education, business, medicine, commerce, economics, and many others.

In this lesson, we will study the terminologies related to testing of hypothesis, how to calculate the probabilities of committing a type I and type II error, hypothesis testing about a population mean when the variance is known, hypothesis testing when the variance is unknown, and hypothesis testing concerning proportions. Hypothesis testing, the focal point lesson, brings to light the role of research in discovering new knowledge and breakthrough in different fields of discipline.

What is It

Example 1. The leader of the association of jeepney drivers claims that the average daily take home pay of all jeepney driver in Navotas City is Php400.00. A random sample of 100 jeepney drivers in Navotas City was interviewed and the average daily take home of these drivers is found to be Php425. Use a 0.05 significant level to find out if the average daily take home pay of all jeepney drivers in Navotas City is different from Php400.00. Assume that the population variance Php8,464.00.

Solution :

A. By critical Value Method

Step 1. State the null and the alternative hypotheses.

H_a: μ =400

H_a: μ ≠400

Step 2 . Choose the level of significance: a= 0.05.

Step 3. Compute the test statistics. Since it is the population mean that is being tested and the population variance is known and n>30, the appropriate test statistic is the z-value.

$$z=\frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$$

Computation:

The standard deviation σ is the square roots of the variance σ^2. The square roots of 8,464 is 92, hence σ = 92.

$$z=\frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}}$$

$$=\frac{425-400}{\frac{92}{\sqrt{100}}}$$

$$=\mathbf{2.72}$$

Step 4. Determine the critical value.

The alternative hypothesis is non- directional; hence the two-tailed test shall be used. Divide α by 2, and then subtract the quotient from 0.5.

$$\frac{\alpha}{2} = 0.025$$

$$0.5 - 0.025 = 0.475 \text{ or } 0.4750$$

Step 5 . Draw a conclusion.

Because the computed test statistic, z= 2.72 falls within the rejection region (beyond the critical value $\pm$ 1.96), reject the null the hypothesis and accept the alternative hypothesis. Conclude that the average daily take home pay of jeepney drivers is not equal to Php400.00. This result is significant at α =0.05 level.

B. By ρ –value method

This method is gaining popularity because of statistical computer programs. Most statistical computer programs are using the p-value method. For deciding and drawing a conclusion., following rules are important.

a. If p-value $\leq \alpha$, reject H_o ,
b. If p-value $> \alpha$, do not reject H_o

Solution

Step 1. State the null and the alternative hypothesis.

H_a: μ =400
H_a: $\mu \neq$400

Step 2. Choose the level of significance: a= 0.05.

Step 3. Compute the test statistics. Since it is the population mean that is being tested and the population variance is known and n>30, the appropriate test statistic is the z-value.

$$\mathbf{z} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Computation:

The standard deviation σ is the square roots of the variance σ^2. The square roots of 8,464 is 92, hence σ = 92.

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

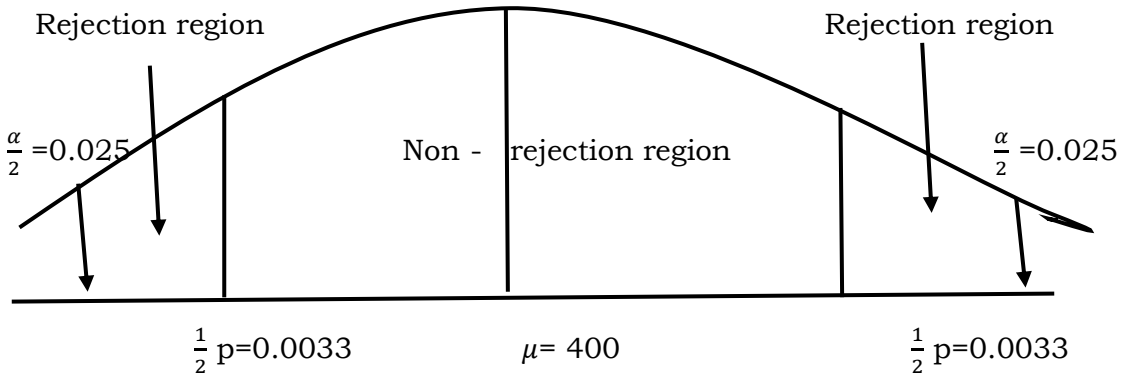
$$= \frac{425 - 400}{\frac{92}{\sqrt{100}}}$$

$$= \mathbf{2.72}$$

Step 4. Determine the critical value

The computed test statistic us z=2.72. Use the Areas under the Standard Normal Curve Table. In the first column under z, look at 2.7. Move to the right along this row until the column headed 2 is reached. The value under the column head 2 is 0.4967. Subtract 0.4967

from 0.5. Since this is a two- tailed test, double the result. Hence, 0.5−0.4967 = .0033 . The p- value = 2 (0.0033) = 0.0066.



5. Draw a conclusion.

Since 0.0066 is less than 0.05, reject the null hypothesis and accept the alternative hypothesis. Conclude that the average daily take home pay of jeepney drivers is not equal to Php400.00. This result is significant at α=0.05 level.

What's More

Directions: Read each problem carefully. Choose the letter which corresponds to the correct answer and write it in a separate sheet of paper.

1- 5. Find the appropriate rejection region in each case. (when the variance is known)

1. H_a: $\mu \neq \mu_o$, α = 0.05.
   a. z=−1.97 b. z=1.95 c. z=1.96 d. z=1.94
2. H_a: $\mu > \mu_o$, α = 0.01.
   a. z=+2.33 b. z=−2.33 c. z= +2.35 d. z= −2.35
3. A two tailed test at 10% level of significance.
   a. The appropriate rejection is the area to the right of the critical value z=+1.645 and the area to the left of the critical value z=−1.645.
   b. The appropriate rejection is the area to the right of the critical value z=−1.645 and the area to the left of the critical value z=+1.645.
   c. The appropriate rejection is the area to the right of the critical value z=+1.635 and the area to the left of the critical value z=-1.635.
   d. The appropriate rejection is the area to the right of the critical value z=+1.545 and the area to the left of the critical value z=-1.545.
4. A two tailed test at 95% level of confidence.
   a. The appropriate rejection is the area to the right of the critical value z=+1.645 and the area to the left of the critical value z=−1.645.
   b. The appropriate rejection is the area to the right of the critical value z=+1.96 and the area to the left of the critical value z=−1.96.
   c. The appropriate rejection is the area to the right of the critical value z=+1.95 and the area to the left of the critical value z=−1.95.

d. The appropriate rejection is the area to the right of the critical value z=+2.96 and the area to the left of the critical value z=-2.96

5. H_a: $\mu < \mu_o$, α = 0.01.
   a. z=2.33 b. −2.33 c. 2.45 d. -2.45

What I Can Do

Test each of the following hypothesis using the given formula.

6. H_o: μ = 84 , H_a: $\mu \neq$ 84. By using the critical value method.
   Given: $\bar{x}$=87, σ =10 , n=35 α =0.05
   a. The computed test statistic z=1.77 does not fall within the rejection region, hence do not reject the null hypothesis.
   b. The computed test statistic z=−1.77 does not fall within the rejection region, hence do not reject the null hypothesis.
   c. The computed test statistic z=1.77 falls within the rejection region, hence, do not reject the null hypothesis.
   d. The computed test statistic z=1.77 does not fall within the rejection region, hence accept the null hypothesis.
7. H_o: μ = 84 , H_a: $\mu \neq$ 84. By using the p- value method.
   Given: $\bar{x}$=87, σ =10 , n=35 α =0.05
   a. The p-value of 0.077 is more than 0.05, hence accept the null hypothesis.
   b. The p-value of 0.077 is less than 0.05, hence do not accept the null hypothesis.
   c. The p-value of 0.077 is more than 0.05, hence do not reject the null hypothesis.
   d. The p-value of 0.077 is less than 0.05, hence accept the null hypothesis.
8. H_o: μ = 45 , H_a: μ < 45. By using the critical value method.
   Given: $\bar{x}$=40, σ =12 , n=32 α =0.01
   a. The computed test statistic z=−2.36 falls within the rejection region, hence reject the null hypothesis.
   b. The computed test statistic z=2.36 do not falls within the rejection region, hence reject the null hypothesis.
   c. The computed test statistic z=2.36 do not falls within the rejection region, hence, accept the null hypothesis.
   d. The computed test statistic z=-2.36 falls within the rejection region, hence, accept the null hypothesis.
9. H_o: μ = 45 , H_a: μ < 45. By using the P- value method.
   Given: $\bar{x}$=40, σ =12 , n=32 α =0.01
   a. The p-value of 0.0091 is less than −0.01, hence reject the null hypothesis.
   b. The p-value of 0.0091 is more than 0.01, hence reject the null hypothesis.
   c. The p-value of 0.0091 is less than 0.01, hence accept the null hypothesis.
   d. The p-value of 0.0091 is less than 0.01, hence reject the null hypothesis.

10- 13. Find the critical value of the following. (When variance is unknown)

10. A right -tailed test, α=0.05 ; df =24.
    a. Critical value =+1.711 c. Critical value = 1.750

b. Critical value = 1.712 d. Critical value = −1.711

11. A left -tailed test; α=0.01 ; df =14.
    a. Critical value =2.553 c. Critical value =−2.553
    b. Critical value =−2.624 d. Critical value= =2.624
12. A two -tailed test, α=0.01; df =18.
    a. Critical value = ±1.734 c. Critical value =−1.734
    b. Critical value = +2.878 d. Critical value = -2.878
13. A two -tailed test, α=0.05; df =16.
    a. Critical value = +2.120 c. Critical value =±2.120
    b. Critical value =−2.120 d. Critical value= 2.120

MODULE 3

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the tests of hypothesis. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module consists of a lesson, namely:

Lesson 1 – Comparing the Sample Mean and the Population Mean in a Large Sample Size

After going through this module, you are expected to:

- Identifies the appropriate rejection region for a given level of significance when:
- the population variance is assumed to be known the population variance is assumed to be unknown the Central Limit Theorem

Lesson 3 — Comparing the Sample Mean and the Population Mean in a Large Sample Size

In this lesson, we shall learn how to determine if a significant difference exists between a sample mean and population mean, using the z-test of one sample mean difference, **we mean that the difference is statistically significant**.

To find out if you are ready to learn this new lesson, do the following activity before going through this lesson.

What is It

Let us explore.

Example 1.

A new drug on the market is claimed by its manufacturers to reduce overweight women by 4.55 kg per month with a standard deviation of 0.91 kg. Ten women chosen at random have reported losing an average of 4.05 kg within a month. Does this data support the claim of the manufacturer at 0.05 level of significance?

For you can easily understand how to test a hypothesis, a simplified approach of testing a hypothesis is presented to you. Understand carefully and suggested to follow it.

I. Problem:	Is the claim true that the drug reduces overweight women by 4.55 kg per month with a standard deviation (σ) of 0.91 kg?
II. Hypotheses:	H_o: The average weight loss per month using a new drug is equal to 4.55 kg ($\mu = 4.55$) H_a = The average weight loss per month using a new drug is not equal to 4.55kg ($\mu \neq 4.55$)
III. Level of Significance: Critical value (cv)	$\alpha = 0.05$ c.v. = ± 2.262
IV. Statistics	t- test for two-tailed test or non-directional test

Example 2:

The ABC company claims that the average lifetime of a certain tire is at least 28 000 km. To check the claim, a taxi company puts 40 of these tires on its taxis and gets a mean lifetime of 25 560 km. With a standard deviation of 1 350 km, is the claim true? Use the z-test at 0.05.

I. Problem	Is the claim true that the average lifetime of a certain tire is at least 28 000 km?
	H_o: The average lifetime of a certain tire is at least 28 000 km.

II. Hypotheses:	(H_o: $\mu \geq 28\ 000$) ***H_a:*** The average lifetime of a certain tire is less 28 000 km. (H_o $\mu < 28\ 000$) Since the claim says that a certain tire is at east 28 000 km. it could also be possible that the alternative hypothesis is **H_a < 28 000**
III. Level of Significance: Critical value (cv)	$\alpha = 0.05$ z = -1.645
IV. Statistics	z-test for one-tailed

What's More

It's your turn.

Determine the decision for each of the following, given the computed and critical value of the z.			
1.	z computed = 1.82	z critical = 1.96	
2.	z computed = 2.54	z critical = 2.33	
3.	z computed = 1.02	z critical = 2.33	
4.	z computed = 2.54	z critical = 2.33	
5.	z computed = 2.54	z critical = 2.33	
Determine the decision for each of the following given the computed z *note:* Determine first the critical value using the confidence level □.			
6.	z computed = 1.29	confidence level= 90% two tailed	
7.	z computed = 1.87	□= 0.05 one tailed	
8.	z computed = 1.11	confidence level= 90% one tailed	
9	z computed = 3.11	□= 0.01 two-tailed	
10.	z computed = 1.34	confidence level = 95% one-tailed	

What I Can Do

Directions: Read and Understand the problem carefully and Solve the following:

A sociologist believes that it costs more than Php 90 000 to raise a child from birth to age one. A random sample of 49 families, each with a child is selected to see if this figure is correct. The average expenses for these families reveal a mean of Php 92 000 with a standard deviation of Php 4 500. Based on these sample data, can it be concluded that the sociologist is correct in his claim? Use the 0.05 level of significance.

I. Problem

II. Hypotheses:
III. Level of Significance: Critical value (cv)
IV. Statistics
Rejection Region:
Compute the test value, using the test statistics
V. Decision Rule:
VI. Conclusion:

A printer manufacturing company claims that its new ink-efficient printer can print an average of 1500 pages of word documents with a standard deviation of 60. Thirty-five (35) of these printers showed a mean of 1 475 pages. Does this support the company's claim? Use the 95% confidence level.

I. Problem
II. Hypotheses:
III. Level of Significance: Critical value (cv)
IV. Statistics
Rejection Region:
Compute the test value, using the test statistics
V. Decision Rule:
VI. Conclusion:

MODULE 4

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Statistics and Probability. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using. This module targets the following learning competencies:

1. Compute for the test statistic value (population mean) (M11/12SP-IVd-1).
2. Draw conclusion about the population mean based on the test-statistic value and the rejection region (M11/12SP-IVd-2).

After going through this module, you are expected to:

- define tests of significance;
- compute the test statistic;
- find the p-value; and
- compare p-value with α; and

draw conclusion about the population mean based on the test-statistic value and the rejection region.

Lesson 4	Tests of Significance

Once a sample data has been collected, researchers will use a tool to find out the probability that a relationship exists between two variables in every sample. They need to assess whether or not the relationship between two variables does exist or it is just because of random chance. In this module, you will learn how to do it. You will know how to compute for the test statistic value (population mean), and draw a conclusion about the population mean. The learning that you gained from the previous modules will help you understand this lesson

What is It

A. Computing for the Test-Statistic Value (Population Mean)

After formulating the null and alternative hypotheses, the next step is to compute the test statistic. However, before doing the computation, you have to identify first the appropriate significance test. Take note that the test statistic follows a normal distribution where the mean is 0 and the standard deviation is 1.

1. Use z-tests when the population standard deviation σ is known.
   This test statistic uses the formula: $z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}$

   where $\bar{X}$ = sample mean
   μ = population mean
   σ = population standard deviation
   n = sample size

2. Use t -tests when the population standard deviation σ is unknown. Actually, this statistical test is more commonly used than a z-test because in most research cases, the population standard deviation σ is not known.
   This test statistic uses the formula: $t = \frac{\bar{X}-\mu}{\frac{s}{\sqrt{n}}}$.

   where $\bar{X}$ = sample mean
   μ = population mean
   s = sample standard deviation
   n = sample size

To summarize when to use a t-test or a z-test, use this diagram:

In the past, statisticians used a z-test when n ≥ 30 and used a t-test when n < 30. That is because they assume that a distribution is normally distributed when the sample

size is large enough. However, there is no need to do it nowadays. We can now use a t-test even if the sample size is greater than or equal to 30. Even the statistical packages now use a t-test for large sample sizes. This is because as the sample size increases, t gets closer to z. Meaning, you do not lose anything when you use a t-test. The main point now is this: **if the population standard deviation (σ) is unknown**, use a t-test **regardless** of the sample size. Meaning, the use of a z-test or a t-test is not related to *n*. So, whenever you use a sample standard deviation (s) to compute the standard error as an **estimate** for a population standard deviation (σ), use a t-statistic.

Example 1: Compute the test statistic using the following data:

$$\bar{X} = 85,\ \mu = 84,\ \sigma = 5,\ n = 60$$

Steps	Solution
1. Identify the appropriate statistical test.	Since the population standard deviation σ is known, use the z-test.
2. Compute using the formula for z statistic	$z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}$ $z = \frac{85-84}{\frac{5}{\sqrt{60}}} = \frac{1}{\frac{5}{7.7460}} = \frac{1}{.6455} =$ z =1.55

Level of significance = 0.05

A = 0.4394 -0.5 = 0.0606 or 6.06%

p-value = 0.0606 > 0.05

Fail to reject the Ho

Example 2: Compute the test statistic using the following data:

$$\bar{X} = 130.05,\ \mu = 120,\ s = 9.96,\ n = 20$$

Steps	Solution
1. Identify the appropriate statistical test.	Since the population standard deviation σ is unknown, use the t-test.
2. Compute using the formula for t statistic	$t = \frac{\bar{X}-\mu}{\frac{s}{\sqrt{n}}}$ $t = \frac{130.05-120}{\frac{9.96}{\sqrt{20}}} = \frac{10.05}{\frac{9.96}{4.4721}} = \frac{10.05}{.6455} =$ 4.512 or 4.51

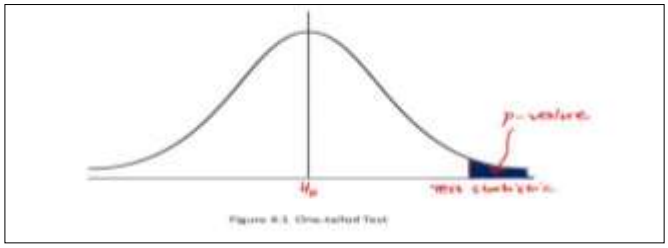
t-value = 4.51, level of significance = 0.05

p-value = 0.00012 < 0.05

Reject the Ho

B. The Probability-value Method (p-Value Method)

Recall that the null hypothesis (H_0) is the claim that is being tested by a test-statistic. You assume this to be true until you have gathered enough evidence that it is not. Once you have found the test statistic, the next step is to find the probability of getting this score when H_0 is true. This probability is known as the *p*-value. The *p*-value approach has become prevalent in testing hypothesis because of the convenience brought to us by computers, calculators, and statistics software.



Modified from Source: https://towardsdatascience.com/how-to-interpret-p-value-with-covid-19-data-edc19e8483b

A *p*-value helps you to determine how likely is the data, assuming that H_0 is true. It is the probability to the right of the test statistic. If you are doing the two-tailed test, then it is the probability to the lower left and to the upper right of the test statistic. Note that it does not tell you the probability that H_0 is true (because in the first place, you assume this to be true before doing the test). This belief is one of the biggest misconceptions about a *p*-value. Another thing is that, having a good *p*-value (or low *p*-value) does not mean that your conclusion is correct. It only tells you how strong your evidence is to reject the null hypothesis. Also, always bear in mind that you do not accept a null hypothesis. It is either you reject it or fail to reject it. This is what we are doing in hypothesis testing. We are gathering evidences to reject the null hypothesis.

Now, how to use the *p*-value in testing the hypothesis?

1. Select the level of significance (α). This is the cutoff value for *p*, and you set this before doing the hypothesis testing. The most commonly used levels of significance are 0.01, 0.05 and 0.10.
2. Compute the *p*-value.
3. Compare the *p*-value with the significance level (α) and draw a relevant conclusion. If the *p*-value is less than or equal to the significance level α, then the evidence is sufficient to reject the null hypothesis.

Interpretation

***p*-value**	**Interpretation**
Less than .01	Highly statistically significant There is very strong evidence against H_0
.01 to .05	Statistically significant Adequate evidence against H_0
Greater than .05	Insufficient evidence against H_0

Adapted from Statistics & Probability by R. Belecina et al, page 259

Decision Rule:

- Reject the null hypothesis when the p-value is equal or smaller than alpha α.
  (Reject H_0 if $p \leq \alpha$)
- Do not reject the null hypothesis when the p-value is larger than alpha α.
  (Do not reject H_0 if $p > \alpha$)

Example:

The owner of a company that sells a particular powdered juice claims that the average weight content of their product is 100 g with a standard deviation of 5 g. However, a group of students wants to test the claim for they believe that it is less than 100 g. So, they get a sample of 50 packs of such powdered juice, computed the weight content, and then find the mean weight to be 99 g. Is the claim of the company owner true?
Solution:

Steps	Solution
1. Formulate the null hypothesis and the alternative hypothesis.	H_0: μ = 100 g / Ho : μ ≥ 100 H_a: μ < 100 g
2. Statistical Test • Choose a significance level (α) • Is the test one-tailed or two-tailed? • What is the appropriate test statistic?	• α = .05 • one-tailed • z test (note that σ is given)
3. Compute for the test statistic and the p-value	$z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}$ $z = \frac{99-100}{\frac{5}{\sqrt{50}}} = \frac{-1}{\frac{5}{7.0711}} = \frac{-1}{.7071} = -1.41$ *The area when z = -1.41 is .4207 p-value = .5000 - .4207 = .0793
4. Compare the p-value with the significance level.	p-value is .0793 α is .05 0.0793 > 0.05
5. Make a conclusion.	Since $p > .05$, the group of students **fail to reject the null hypothesis**, and the result is not significant at $p < .05$. In context, the group of students does not have enough evidence that the weight content of each pack is less than 100 g.

Note:

For a t statistic, it is better to use software or Excel to find the exact p-value. However, if you need to find the p-value manually, you may use a t-table and approximate the probability.

What's More

Activity: Rejected or Not?

Directions: Complete the table by filling out the missing values. Then, draw a decision about the population mean based on the test statistic value and the probability value. (*Assume that there is only one variable and that all the assumptions are met.*)

	Significance Level	Test Statistic (one-tailed)	*p*-value	Decision (Reject the null hypothesis or failed to reject the null hypothesis)
1	$\alpha = .05$	z = 1.35		
2	$\alpha = .10$	z = -2.28		
3	$\alpha = .01$	z = -1.17		
4	$\alpha = .05$	z = 1.96		
5	$\alpha = .05$	z = 2.54		
6	$\alpha = .01$	t = 1.345; n = 15		
7	$\alpha = .10$	t = -1.19; n = 5		
8	$\alpha = .05$	t = 2.756; n = 30		
9	$\alpha = .01$	t = 3.25; n = 10		
10	$\alpha = .01$	t = -1.059; n = 25		

What I Can Do

Directions:

1. Compute the test statistic using the appropriate statistical test. (Write the test statistic in three-digit form.)
2. Find the *p*-value.
3. Using the selected significance level, decide whether to reject the null hypothesis.

$\bar{x} = 102,\ \mu = 100,\ \sigma = 5,\ n = 36,\ \alpha = .05,$ one-tailed

$\bar{x} = 48.95,\ \mu = 50,\ s = 5,\ n = 25,\ \alpha = .05,$ one-tailed test

$\bar{x} = 24.8,\ \mu = 25,\ s = 5,\ n = 25,\ \alpha = .05,$ one-tailed test

MODULE 5

What I Need to Know

This module was designed and written with you in mind. It is here to help you learn about Hypothesis testing. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of learners. The lessons are arranged to follow the standard sequence of the course.

The module is divided into 2 lessons, namely:

- Lesson 1 – Testing Hypothesis involving Population Mean
- Lesson 2 – Testing Hypothesis involving Population Proportion

After going through this module, you are expected to:

1. Solves problems involving test of hypothesis on the population mean.
2. Formulates the appropriate null and alternative hypotheses on a population proportion.
3. Solve problems involving hypotheses on a population proportion.

Lesson 5.1 Testing Hypothesis involving Population Mean

The hypothesis or claims about population mean or population proportion could be tested using the five -step hypothesis testing procedure. There are certain situations when the data to be analyzed involved population proportion or percentage.

What is It

Solves problems involving test of hypothesis on the population mean.

Example:

The owner of the iPhone 12 pro claims that their cellphone has 2,185 mAh Battery with a standard deviation of 60. Forty-five (45) of theses cellphones showed a mean of 2,160 mAh battery. Does this support the company's claim? Use 95% confidence Level.

Answer

Using the five-step hypothesis testing procedure:

1. Null Hypothesis (H_0) and Alternative Hypothesis (H_a)

H_0: $\mu = 2\ 185$ H_a: $\mu \neq 2\ 185$

2. Statistical Test

Since n=45, therefore it is Z-test
We are using equal/not equal sign, it is two-tailed
Confidence Level = 95%, α=0.05

Z-Critical = ±**1.96**

3. Computation

$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$	Write the appropriate equation, by looking at the standard deviation and sample size
$z = \frac{2160 - 2185}{60 / \sqrt{45}}$	Evaluate the given
$z = \frac{-25}{8.944272}$	Perform the operation. Write at least 6 decimal places.
z = -2.795084 = -2.80	The answer is **2.80** (negative sign will be disregarded since the test is two-tailed)

4. Decision (Reject or Not Reject the H_o)

Z-Computed Z-Critical

2.80 > 1.96 -------> **H_0 is Rejected**

5. Conclusion

There is a enough evidence to deny the owner's claims.

Lesson 5.2 Testing Hypothesis involving Population Proportion

The Population Proportion can be estimated only for large sample size (n ≥ 30). The same is true in testing a claim or hypothesis about the population proportion (p).

For example, IATF (The Inter-Agency Task Force on Emerging Infectious Diseases is a task force organized by the executive of the Philippine government to respond to affairs concerning emerging infectious diseases in the Philippine) is studying on the rapid growth of COVID-19 Patients in a region, to determine the proportion of COVID-19 female patients. They don't need to collect all the patient, but they only need a sufficient sample from which they will make inference about the proportion of COVID-19 female patients.

In the example above, AITF may initially believe that 50% of the patients are female. Suppose they gather enough data. Out of 100 records, 56 are female patients. Would this support their initial belief?

To test a claim about population proportion, we use the ***z*-test** for **Population**

Proportion. The formula below is used.

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}}$$

Where:

p = claimed / hypothesis proportion

p̂ = sample proportion (p̂ =x/n)

q = 1 - p

n = sample size

as in the use of the z-test for means, the decision rule below is used:

Z-computed ≥ Z-critical -----> **Reject H_0**

Z-computed < Z-critical -----> **Do not Reject / Accept H_0**

Example: Compute for z for each of the following given.

1. p = 0.30, p̂ = 0.40 , n= 30
2. p = 0.87, p̂ = 0.81 , n= 45

Answer: 1.

q = 1 – p = 1 – 0.30 = 0.70	Compute for the value of q
$z = \frac{\hat{p} - p}{\sqrt{pq/n}}$	Write the equation
$z = \frac{0.40 - 0.30}{\sqrt{(0.30)(0.70)/30}}$	Evaluate the given
$z = \frac{.10}{.083666}$	Perform mathematical operation (GEMDAS), Write at least 6 decimal places.
Z= 1.195229	The answer is **1.2**

q = 1 – p = 1 – 0.87 = 0.13	Compute for the value of q
$z = \frac{\hat{p} - p}{\sqrt{pq/n}}$	Write the equation
$z = \frac{0.81 - 0.87}{\sqrt{(0.87)(0.13)/45}}$	Evaluate the given
$z = \frac{-0.06}{.050133}$	Perform mathematical operation (GEMDAS) Write at least 6 decimal places.
Z= -1.196813	The answer is **-1.2**
$z = \frac{0.033846}{.028501}$	Perform mathematical operation (GEMDAS) Write at least 6 decimal places.
Z= 1.187537 = 1.19	The answer is **1.19**

What's More

Determine the decision for each of the following given.
Write R if Rejected, DNR if Do not Reject, the Null hypothesis.

1. Z-computed= 2.25 Z-critical=2.87
2. Z-computed= 1.95 Z-critical=2.50
3. Z-computed= 0.89 Z-critical= 0.89
4. Z-computed= 1.00 Z-critical= 3.00
5. T-computed= 0.27 T-critical= 3.00
6. T-computed= 1.56 T-critical= 1.97
7. T-computed= 2.34 T-critical=2.43
8. T-computed= 1.23 T-critical=2.13
9. Z-computed= 0.12 Confidence level=90% , one tailed
10. Z-computed= 1.97 Confidence level= 95% , two tailed
11. Z-computed= 2.22 α= 0.01 , one tailed
12. T-computed= 1.11 Confidence level= 95% , two tailed, n=18
13. T-computed= 1.67 α =0.1 , one tailed, n=20
14. T-computed= 1.67 α =0.1 , Two tailed, n=20
15. T-computed= 2.50 α = 0.05 , one tailed, n= 1

What I Can Do

Answer the given questions.

1. In a recent survey, a researcher claims that the average life of a dog in a certain country is 10 years. Is their claim correct if a random sample of 30 deaths from this country showed a mean of 13 years with a standard deviation of 1.2 years? Use 95% confidence level.

2. Ms. Pelaez, a teacher in English, believes that less than 15% of the student like English, If 20 out of 55 randomly students like English, is the teacher's claim valid? Use 95% confidence level.

MODULE 6

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of Statistics and Probability. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using. The module consists of one lesson which contains sub lessons:

- Lesson 6 – Comparing Sample Proportion and Population Proportion

Lesson 6	Comparing Sample Proportion and Population Proportion

In daily life, we make tentative explanation of facts about a particular phenomenon by formulating hypothesis. This hypothesis may be correct or incorrect, depending on the available evidence that we can gather to support our hypothesis. We usually use a sample to gather information and evidence that we need to validate our hypothesis. The data that we gather from this sample become the basis of our decision whether we shall accept or reject our hypothesis regarding the entire population. The data obtained from this sample is analyzed with the use of appropriate statistical procedure to find out whether our hypothesis should be accepted. This process is called testing hypothesis. In this lesson we shall learn how to determine if a proportion from a sample differs significantly from a proportion from a population. We shall learn how to conduct a test of hypothesis that will help us to arrive at the right decision. There are some instances wherein what we want to compare are proportions.

What is It

To compare sample proportion and population proportion, we use the z-test for one-sample proportion. The test statistics for this test is

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \qquad z = \frac{\hat{p} - p}{\sqrt{\frac{pq}{n}}}$$

where;

$\hat{p}$ = sample proportion $\qquad \hat{p} = \frac{x}{n}$

p_o = population proportion $\qquad$ x = number of successes

n = size of the sample

Example 3

It has been claimed that less than 60% of all purchases of a certain kind of computer program will call the manufacturer's hotline within one-month purchase. If 55 out of 100 software purchasers selected at random call the hotline within a month of purchase, test the claim at 0.05 level of significance.

Solution

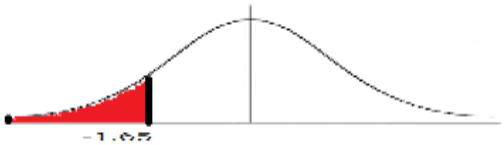
Step 1: Formulate the null and alternative hypotheses

H_o: The proportion of purchasers that will all the manufacturer's hotline within one month of purchase is 60% or 0.60

($p_o \geq 0.60$)

H_a: The proportion of purchasers that will call the manufacturer's hotline within one month of purchase is less than 60% or 0.60 **(p_o < 0.60)**

Step 2: Type of test: The test is one-tailed (left tailed).
Critical value: with the use of z-table,
the critical value of z at 0.05 level,
One-tailed test is z =-1.65.



Step 3: Compute the test value.

Given:

$p_o = 0.60$

$n = 100$

$\hat{p} = \frac{55}{100} = 0.55$

Substitute the given values in the formula below.

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$z = \frac{0.55 - 0.60}{\sqrt{\frac{0.60(1 - 0.60)}{100}}}$$

$$z = \frac{-0.05}{\sqrt{\frac{0.60(0.40)}{100}}}$$

$$z = \frac{-0.05}{\sqrt{\frac{0.24}{100}}}$$

$$z = \frac{-0.05}{\sqrt{0.0024}}$$

$\sqrt{0.0024} = 0.04898 = 0.0490$ round off to the nearest ten thousandths

$$z = \frac{-0.05}{0.0490}$$

z = -1.02

Step 4: Decision:

Fail to reject /Accept the null hypothesis because the computed value or the test value falls outside the rejection region.

Step 5: Conclusion:

There is no sufficient evidence to conclude that the proportion of purchasers that will call the manufacturer's hotline within one month of purchase is less than 60%. Thus, the claim is false or incorrect.

What's More

Solve Me Please!

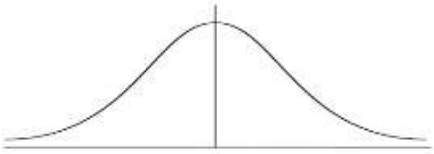
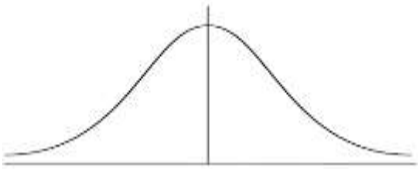
Directions: Analyze and solve the given problem below.

A doctor claims that only 10% of all patients exposed to a certain amount of radiation will feel ill effects. If in a random sample, 5 of 18 patients exposed to such radiation feel some ill effects, test the doctor's claim at 0.01 level of significance.
1. Formulate the null and alternative hypotheses H_o: H_a:
2. Type of test:
3. Compute the test value.
4. Decision:
5. Conclusion:

What I Can Do

What's My Decision?

Directions: Find the critical value, type of test, draw the rejection region, compute the value of the test statistic, and make a decision whether to accept or failed to accept the null hypothesis in each of the following situations.

Hypotheses	Rejection Region	Decision
1. H_0: $p_o \leq 0.58$ H_a: $p_o > 0.58$ Given: $p_o = 0.58$ $x = 80$ $n = 120$ $\alpha = 0.05$		
Critical value:	Test statistic:	
Type of test:		
2. H_0: $p_o \geq 0.80$ H_a: $p_o < 0.80$ Given: $p_o = 0.80$ $x = 140$ $n = 200$ $\alpha = 0.01$		
Critical value:	Test statistic:	
Type of test		

Assessment

Directions: Choose the letter of the correct answer. Write your answer on a separate sheet of paper.

1. Is a numerical quantity that is assigned to the outcome of an experiment.
   A. Random variable
   B. Sample point
   C. Sample space
   D. Variable
2. In how many ways can two coins fall?
   A. 2 B. 4 C. 6 D. 8
3. It tells the distance of score from the mean measured in standard deviation units.
   A. normal curve
   B. sample mean
   C. z-score
   D. area
4. Which of the following shows the probability that the z-score lies above a z-score value?
   $A. P(a < z < b)$ $B. P(z > a)$ C. $P(z < a)$ D. $P(a = z$

5. What is the proportion of the area to the right of z = -1?
   A. -0.3413 B. -0.8413 C. 0.3413 D. 0.8413

6. **Statement 1**: The number of students who are present in Filemon T. Lizan SHS for the first day of class for the S.Y. 2020-2021
   Statement 2: The number of Mayors in NCR who are present during the meeting
   Which of the following is CORRECT?
   A. both statements are Discrete Random Variables
   B. both statements are Continuous Random Variables
   C. Statement 1 is Discrete Random Variable while the Statement 2 is Continuous Random Variables
   D. Statement 1 is a Continuous Random Variable while the Statement 2 is a Continuous Random Variable
7. Statement 1: the volume of soft drinks in a 12-ounce can
   Statement 2: the time required to perform a job.
   which of the following is CORRECT?
   A. both statements are Discrete Random Variables
   B. both statements are Continuous Random Variables
   C. Statement 1 is Discrete Random Variable while the Statement 2 is Continuous Random Variables
   D. Statement 1 is a Continuous Random Variable while the Statement 2 is a Continuous Random Variable
8. Let B number of boys in a family and G for the girls in a family of four children. Determine the values of the random variable B.
   A. 0, 1 B. 0, 1, 2 C. 0, 1, 2, 3 D. 0, 1, 2, 3, 4

For numbers 9 – 10. Consider the probability distribution of the number of mangoes given below.

R	3	2	1	0
P(R)	1/8	3/8	3/8	1/8

9. Find P(R = 3) A. 1/8 B. 5/8 C. 3/8 D. 1
10. Find P(R > 1)
    A. 18 B. 3/8 C. 1/2 D. 1

Answer Key

What's More

1. DRV 11. DRV
2. CRV 12. DRV
3. DRV 13. CRV
4. CRV 14. CRV
5. CRV 15. CRV
6. CRV 16. DRV
7. DRV 17. DRV
8. CRV 18. CRV
9. CRV 19. CRV
10. DRV 20. CRV

What I Can Do?

1. a. 16 elements
b. 4,3,2,1,0
c. 4, 3, 2,1,0 or 0,1,2,3,4
2. a. 7 elements
b. 3,2,1
c. 3,2,1 or 1,2,3
3. a. 7 elements
b. 3,2,1
c. 3,2,1 or 1,2,3

What I Know

1.	A	11. C	
2.	B	12. A	
3.	C	13. A	
4.	B	14. B	
5.	D	15. B	
6.	A	16. B	
7.	B	17. A	
8.	D	18. B	
9.	A	19. C	
10.	C	20. B	

What's More

1. $z = -2.24$
2. $z = 1.24$
3. $z = -2.74$
4. $z = -0.75$
5. $z = 3.35$

What I Can Do?

1. $z = -2.40$
Area = 0.0082 or 0.82%
2. $z = -1.67$
Area = 0.9525 or 95.25%

What's More

1. $\mu = 0.08$
2. $\sigma = 1.711$

What I Can Do?

What's More

1. $\mu = 2.8$
2. $\sigma^2_X = 1.25$
3. $\sigma_X = 1.118$ or 1.12
4. $\mu_X = 48$
5. $\sigma^2_X = 1.4083$ or 1.41
6. $\sigma_X = 1.1867$ or 1.19
7. $\mu_X = 120$
8. $\sigma^2_X = 6.5333$
9. $\sigma_X = 2.5560$ or 2.56

What I Can Do?

10. $\mu_X = 42$
11. $\sigma^2_X = 2.8648$ or 2.86
12. $\sigma_X = 1.6926$ or 1.69
13. $\mu_X = 42$
14. $\sigma^2_X = 3.1322$ or 3.13
15. $\sigma_X = 1.7698$ or 1.77

What's More

1. Independence
2. equal
3. ten
4. zero
5. larger
6. normal
7. Sample size
8. random
9. mean
10. CLT

Across: 2, 6, 8
Down: 1, 3, 4, 5, 7, 9

What I Can Do?

Performance Task

What's More

The Neophyte Statistician
703 children

What I Can Do?

1. 80
2. 0%
3. 305

What's More

1. 3
2. 10
3. 21
4. 36
5. 10
6. 21

What I Can Do?

a. $\$21$
b. 8 or possible sample
c. construct sampling dist. Of the sample means
d. construct the histogram

What's More

1. $\mu = \frac{20}{6}$ or 3.5
2. $\mu = \frac{19}{10}$ or 1.9

What I Can Do?

1. $\mu = 2$
2. $\mu = 1.23$
3. $\sigma^2 = 2$
4. $\sigma = 1.4142$
5. $\sigma = 0.7628$

What's More

1. $\mu = 2$
2. $\mu = 1.23$
3. $\sigma^2 = 0.5819$
4. $\sigma = 0.7628$

What's More

x	2	3	4	5	6	7
P(x)	0.15	0.20	0.25	0.20	0.20	0.20

x	2	3	4	5	6
P(x)	0.25	0.12	0.25	0.18	0.15

x	1	2	3	4
P(x)	0.49	0.20	0.18	0.10

x	1/2	1/2	1/2	1/2	1/2	1/2
P(x)	1/2	1/2	1/2	1/2	1/2	1/2

What I Can Do?

A. B.
1. 124 1. $\sigma^2 = 0.0875$
2. 147 2. $\sigma = 0.9296$
3. 148 3. $\sigma^2 = 0.0208$
4. 144 4. $\sigma = 0.9165$
5. 139

Quarter Module 1

Module 2

Module 3

Module 4

Module 5

Module 6

Module 7

Module 7

Module 8

Module 9

Module 10

What's More

A.
1. 0.25
2. 0.426
3. 0.224
4. 0.45
5. 0.222
B.
6. 0.226
7. 0.282
8. 0.23
9. 0.242
10. 0.264

What I Can Do?

11. 0.093 or 0.09
12. 0.1042 or 0.10
13. 0.0967 or 0.10
14. 0.0956 or 0.10
15. 0.1117 or 0.11
16. 0.3267 or 0.33
17. 0.196 or 0.20
18. 0.3222 or 0.32
19. 6.0615 or 6.06
20. 7.6845 or 7.68

Assessment

1. B
2. C
3. D
4. B
5. A
6. C
7. C
8. A
9. B
10. A

What's More

1. p = 0.10 or 10%
   P≠ 0.10 or 10%
2. t-test, t = ±2.898
3. t= 2.55
4. Accept/Fail to Reject H_0
5. There is a significant difference between the sample mean and population mean. Thus, there is enough evidence to accept the doctor's claim

What I Can Do?

1. critical value: z = 1.645
   type of test: one-tailed (right tailed)
   test statistic: z = 1.9975 or 2
   Decision: Reject the H_0

2. critical value: z = -2.326
   type of test: one-tailed (left tailed)
   test statistic: z = -3.5355 of -3.54
   Decision: Reject the H_0

Module 5

What's More

1. z=0.069, Accept/Fail to Reject H_0
2. z=0.011, Reject H_0
3. z=0.121, Accept/Fail to Reject H_0
4. z=0.025, Reject H_0
5. z=0.006, Reject H_0
6. t=0.10, Accept/Fail to Reject H_0
7. t=0.15, Accept/Fail to Reject H_0
8. t=0.005, Reject H_0
9. t=0.005, Reject H_0
10. t=0.19, Accept/Fail to Reject H_0

What I Can Do?

I.
a. z = 2.4
b. p-value = 0.008
c. Reject H_0
2.
a. t = -1.05
b. p-value = 0.152
c. Accept/Fail to Reject H_0
3.
a. t = -0.20
b. p-value = 0.422 c. Accept/Fail to Reject H_0
c. Reject H_0

Module 4

What's More

1.	DNR	11.DNR
2.	DNR	12. DNR
3.	R	13. R
4.	DNR	14. DNR
5.	DNR	15. R
6.	DNR	
7.	DNR	
8.	DNR	
9.	DNR	
10.	R	

What I Can Do?

1. z = 13.69
   Reject the H_0
2. z = 4.36
   Reject the H_0

Module 6

What's More

1. Accept/Fail to Reject H_0
2. Reject H_0
3. Accept/Fail to Reject H_0
4. Reject H_0
5. Reject H_0
6. ±1.645 Accept/Fail to Reject H_0
7. 1.645 Reject H_0
8. 1.282 Accept/Fail to Reject H_0
9. ±2.576 Reject H_0
10. 1.645 Accept/Fail to Reject H_0

What I Can Do?

1.
I. Is the claim true that it costs more than Php90 000 to raise a child from birth to age one?
II. H_0: $\mu \leq Php90\ 000$ H_a: $\mu > Php90\ 000$
III. 0.05, 1.645
IV. z-test for one-tailed, z = 3.11
V. Reject the H_0
VI. There is a significant difference between the sample mean and population mean. Thus, the sociologist is correct in claiming that the cost to raise a child from birth to age one is more than Php90 000.

Module 3

What's More

1. C
2. A
3. A
4. B
5. B

What I Can Do?

6. A or D
7. A or C
8. A
9. D
10. A
11. B
12. B
13. C

Module 2

What's More

1. Null Hypothesis
2. Alternative Hypothesis
3. Alternative Hypothesis
4. Null Hypothesis
5. Alternative Hypothesis
6. Null Hypothesis
7. Null Hypothesis
8. Null Hypothesis
9. Null Hypothesis
10. Alternative Hypothesis

What I Can Do?

1.two-tailed	1. Reject H_0
2.one-tailed	2. Accept/Fail to Reject H_0
3.one-tailed	3. Reject H_0
4. one-tailed	4. Reject H_0
5. two-tailed	5. Reject H_0
6. two-tailed	6. Accept/Fail to Reject H_0 7. one-tailed
7. Accept/Fail to Reject H_0	
8. one-tailed	8. Reject H_0
9. one-tailed	9. Reject H_0
10. two-tailed	10. Reject H_0

Quarter 2 Module 1

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